Marine Microbial Biodiversity

Lecture 08

Eukaryotes and Viruses diversity in Marine Environments

Diversity 3

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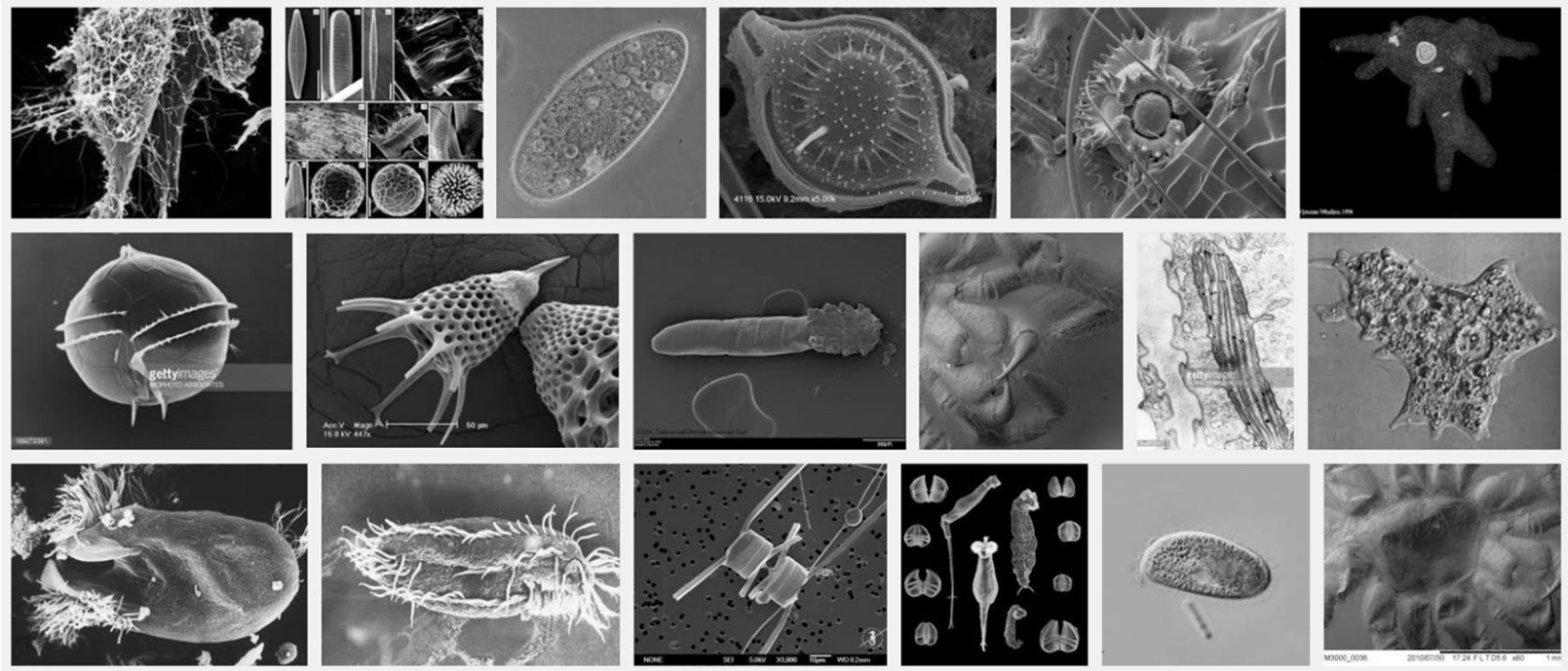






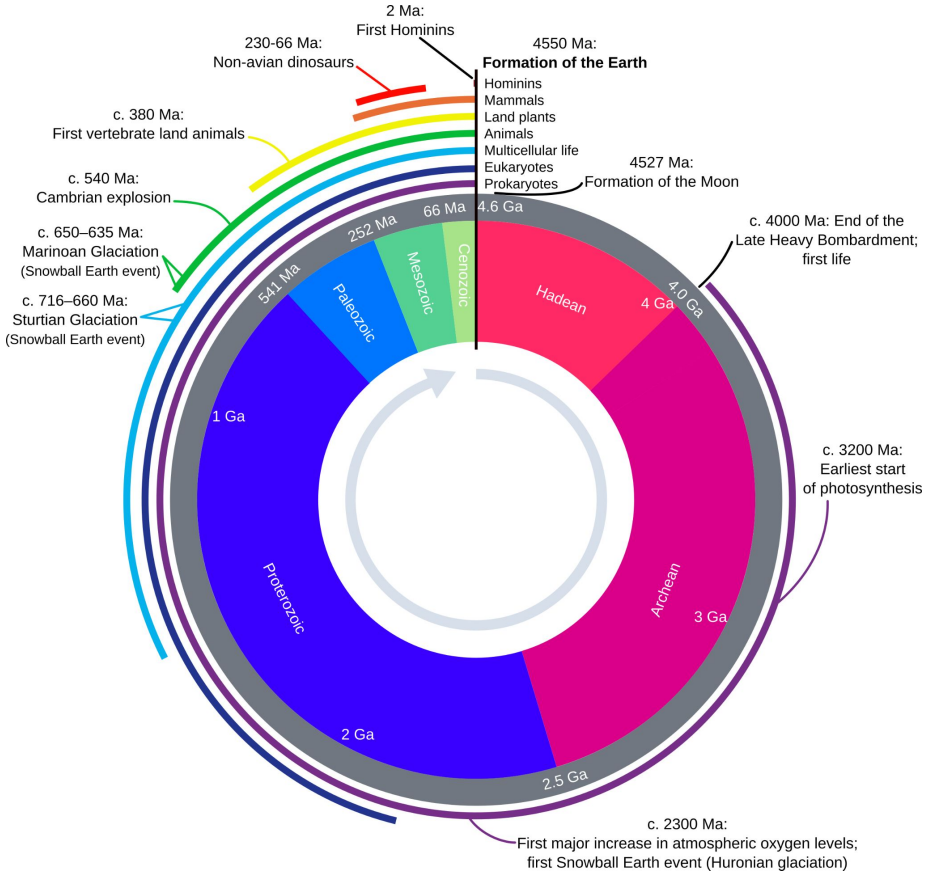
Microbial Eukaryotes

Microbial eukaryotes are microscopic, single-celled or simple multicellular organisms whose cells contain a membrane-bound nucleus and other specialized, membrane-bound internal structures (organelles)



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Origin And Evolution Of Life On Earth



Origin of Earth: 4.55 ± 0.07 Ga

LUCA: 4.52 - 4.48 Ga

Earliest fossil: 4.28 - 3.77 Ga

Anoxygenic photosynthesis: 3.47 - 3.33 Ga

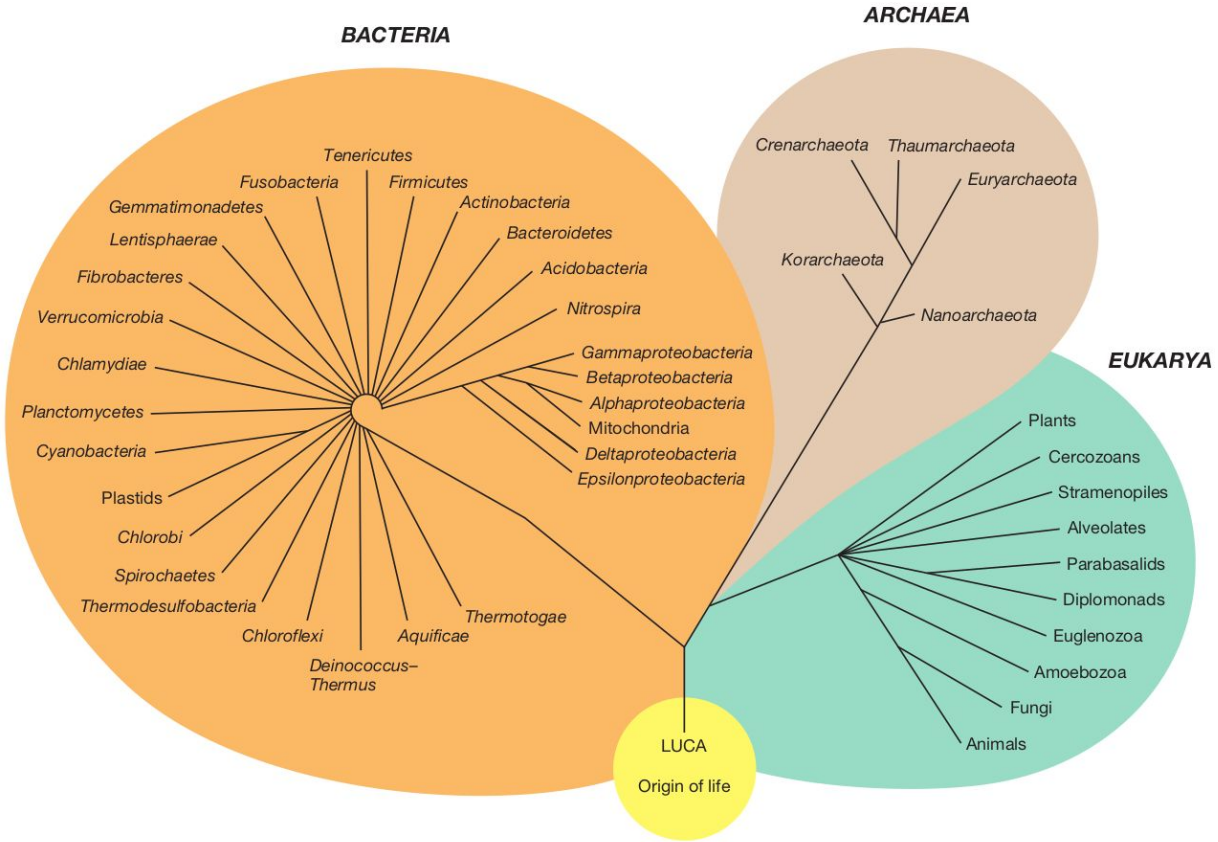
Oxygenic photosynthesis : 2.8 Ga

GOE: 2.45 - 2.32 Ga

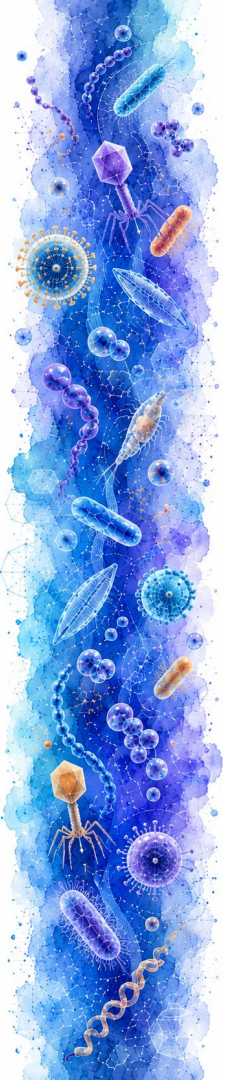
Earliest eukaryotic fossil: 2.33 Ga

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The Universal Tree Of Life Based On rRNA Gene Sequences



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How Did Eukaryotes Form?

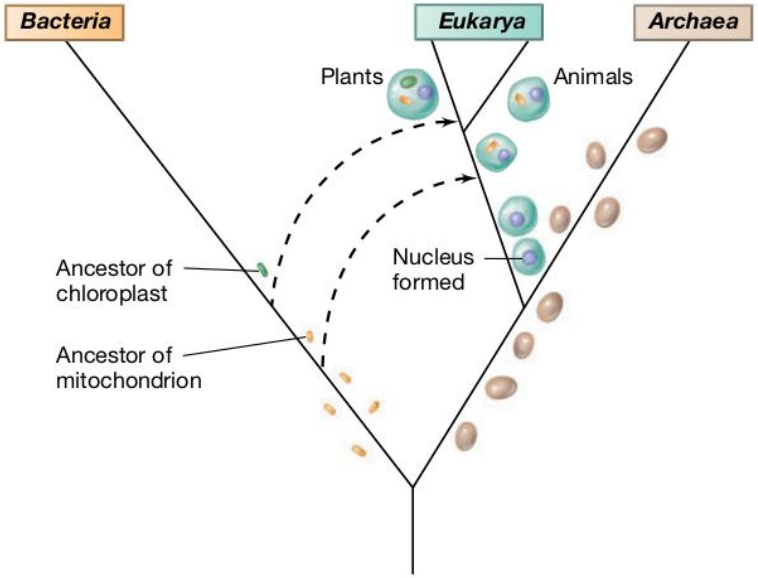
Symbiogenesis, or **endosymbiotic** theory, is a theory on the evolution of eukaryotes, first hypothysed by the Russian botanist Konstantin Mereschkowski in 1905 and then proposed and supported by American evolutionary biologist Lynn Margulis in 1967.

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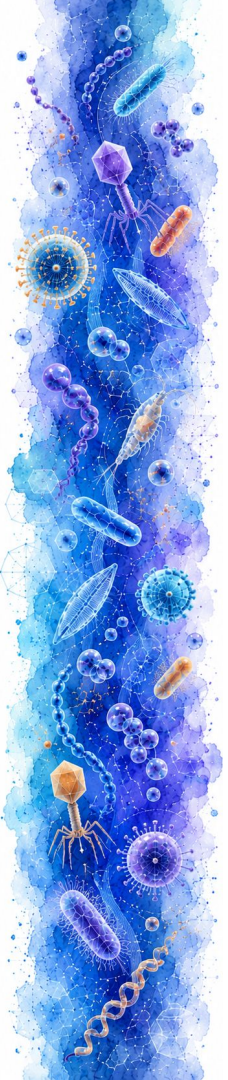
Endosymbiosis And Eukaryotic Cell Evolution

Symbiogenesis, or **endosymbiotic theory**, is a theory on the evolution of eukaryotes, first hypothysed by the Russian botanist Konstantin **Mereschkowski** in 1905 and then proposed and supported by American evolutionary biologist Lynn Margulis in 1967.

Endosymbiosis is the process in which a ancient archaeal host cell engulfing a bacterium that, rather than being destroyed, managed to survive and replicated as an endosymbiont within the cytoplasm of the host cell. The mitochondria and chloroplasts are thought to represent formerly free-living bacteria incorporated in an endosymbiotic event around 2 billion years ago. Molecular and biochemical evidence suggest that **mitochondria** developed from **Proteobacteria** (Alphaproteobacteria, like SAR11) and **chloroplasts** from **Cyanobacteria** (in particular, nitrogen-fixing filamentous cyanobacteria).



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Support For The Endosymbiotic Hypothesis

.Mitochondria and chloroplasts can arise only from preexisting mitochondria and chloroplasts. They cannot be formed in a cell that lacks them because nuclear genes encode only some of the proteins of which they are made

.Both have their own genome, and it resembles that of bacteria not that of the nuclear genome. Both genomes consist of a single circular molecule of DNA with no associated histones (only NAPs)

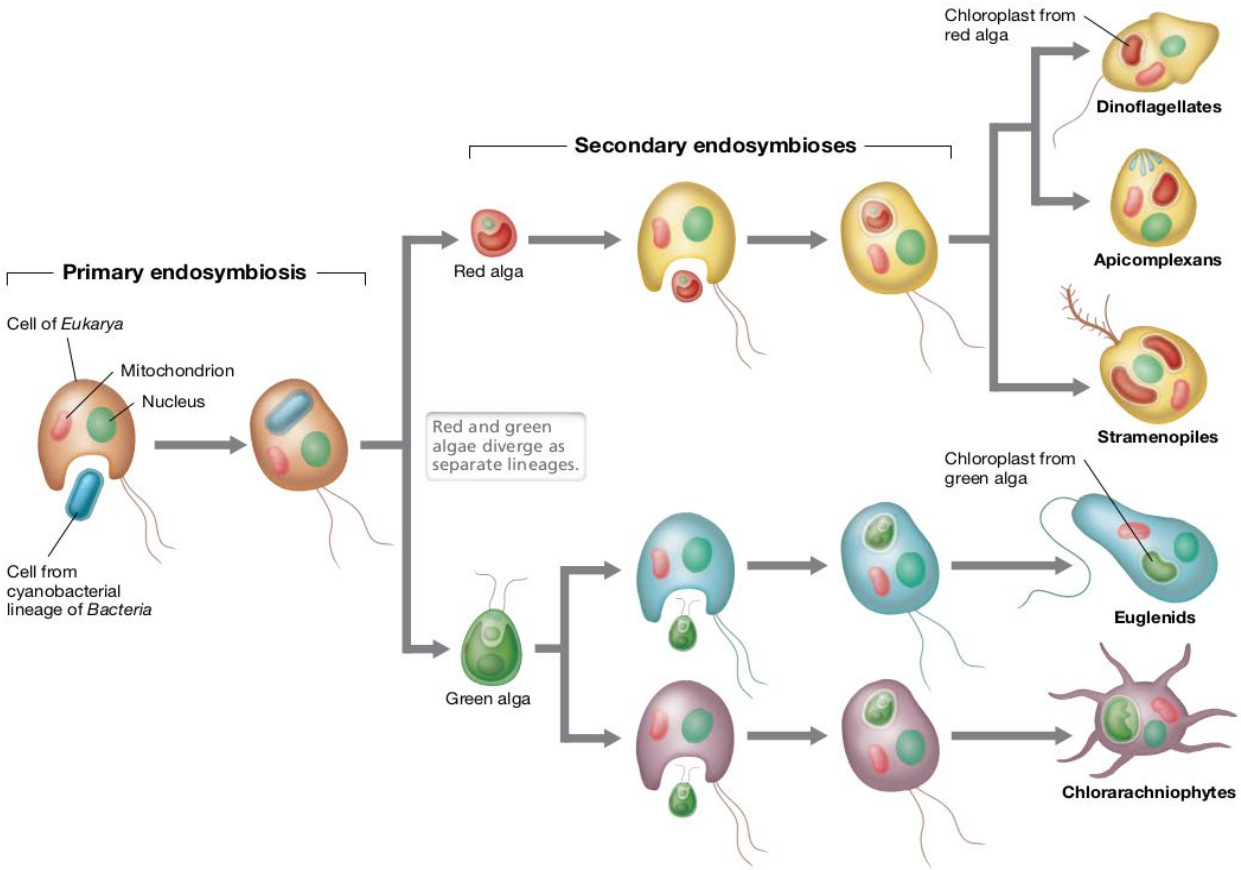
.Both mitochondria and chloroplasts have their own protein-synthesizing machinery, and it more closely resembles that of bacteria containing 70S ribosomes, rather than the 80S, similar in size and structure to bacterial ribosomes, that is the reason why they are sensitive to a number of antibiotics (e.g. streptomycin, tetracycline and those acting by interfering with protein synthesis)

.Both mitochondria and chloroplasts are surrounded by double membranes. The inner membrane is thought to resemble the plasma membrane of the ancestral bacterial cell, while the outer membrane resembles that of the host eukaryotic cell

.During cell division, these organelles divide autonomously within the cell, through a process similar to binary fission, rather than being assembled from scratch, supporting the idea of a self-replicating ancestral origin

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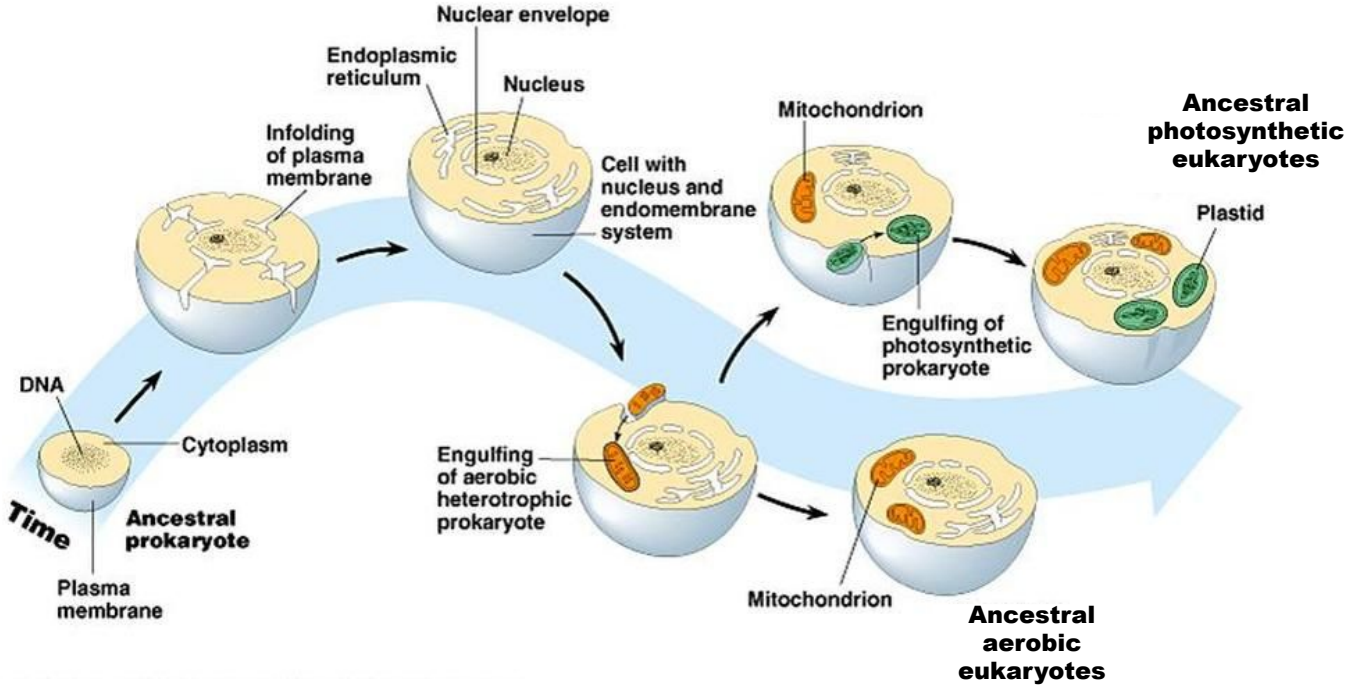
Primary And Secondary Endosymbiosis



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Primary And Secondary Endosymbiosis

Primary endosymbiosis refers to the original internalization of prokaryotes by an ancestral a proto-eukaryote, deriving from the archaeal lineage, resulting in the formation of the mitochondria and chloroplasts. Two primary endosymbiotic events are central to the **evolution of aerobic** eukaryotes and **photosynthetic eukaryotes**.



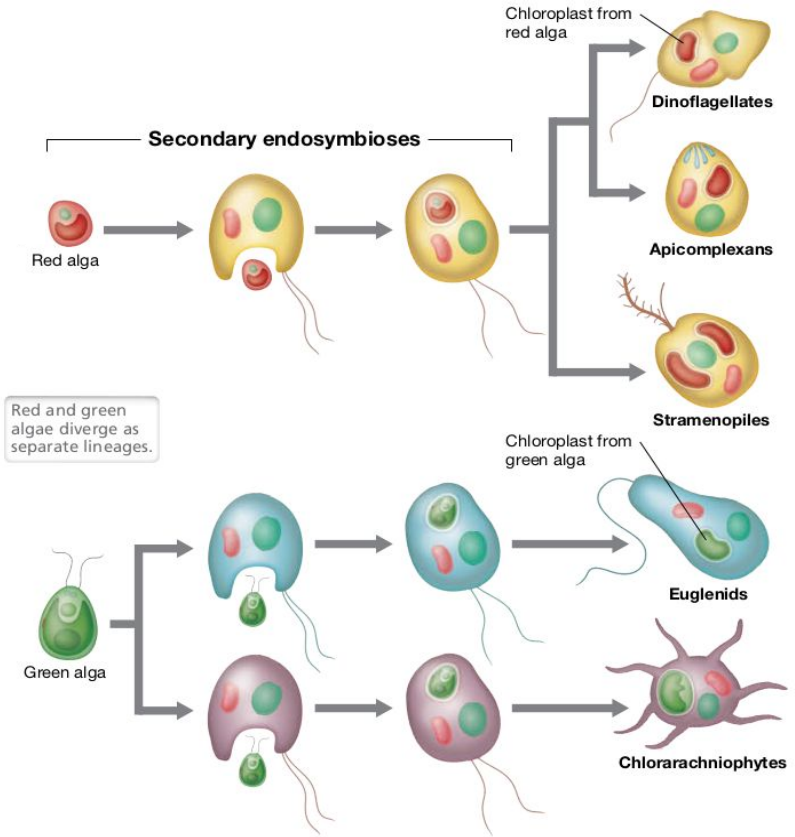
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Primary And Secondary Endosymbiosis

Secondary endosymbiosis occurs when a eukaryotic cell engulfs another eukaryotic cell that already contains a primary endosymbiont, usually a photosynthetic eukaryote (like a green or red alga with a chloroplast from primary endosymbiosis). The engulfed cell eventually becomes an organelle within the new host cell, leading to further complexity in cell structure and metabolism.

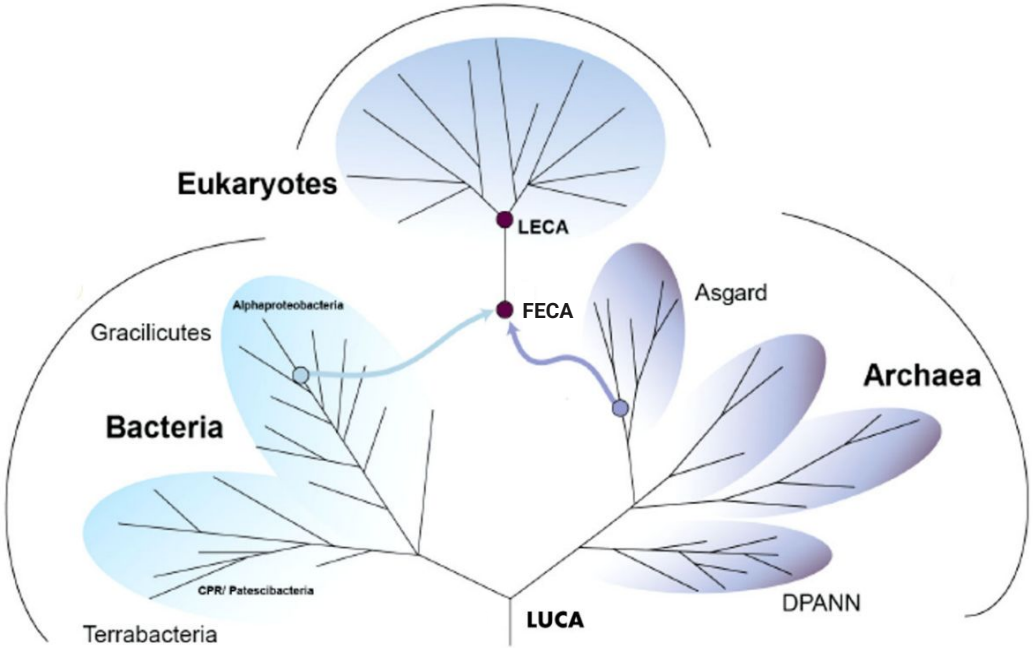
Secondary endosymbiosis involving red algae led to the chloroplasts found in groups like dinoflagellates, diatoms, brown algae, and apicomplexans.

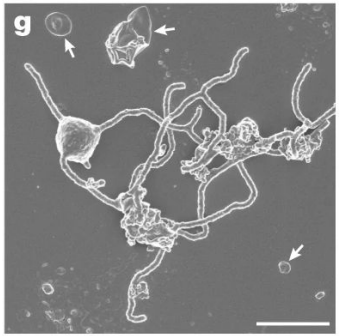
Secondary endosymbiosis with green algae gave rise to chloroplasts in groups such as **euglenoids** and **chlorarachniophytes**.



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FECA And LECA





Lokiarchaeota

First Eukaryotic Common Ancestor (FECA) and Last Eukaryotic Common Ancestor (LECA) are two hypothesized stages in the early evolution of eukaryotes, describing significant evolutionary transitions on the path from prokaryotic-like ancestors to modern eukaryotic cells.

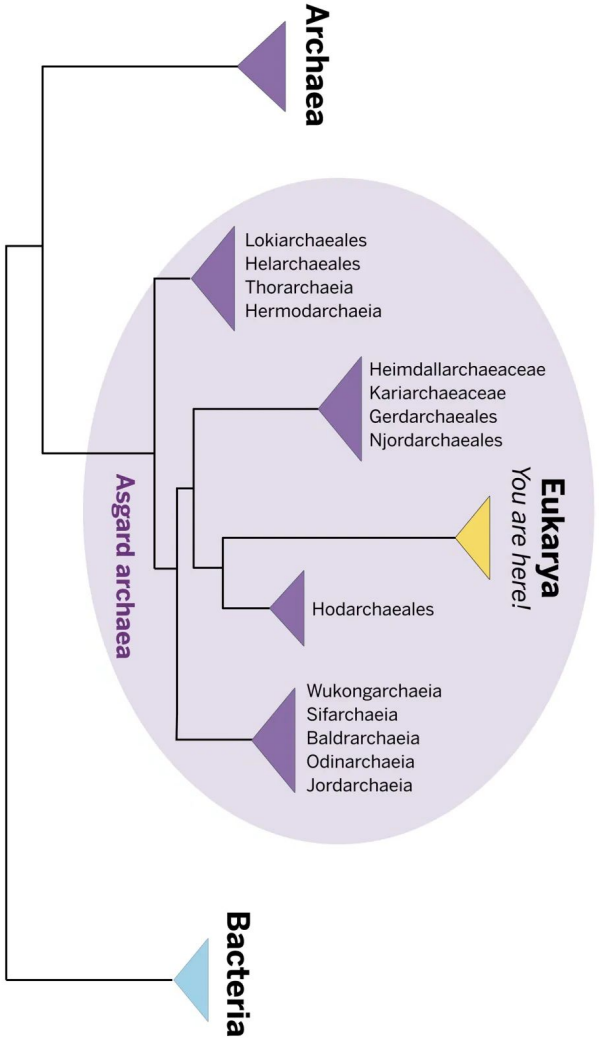
https://doi.org/10.1038/s41586-019-1916-6

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FECA And LECA

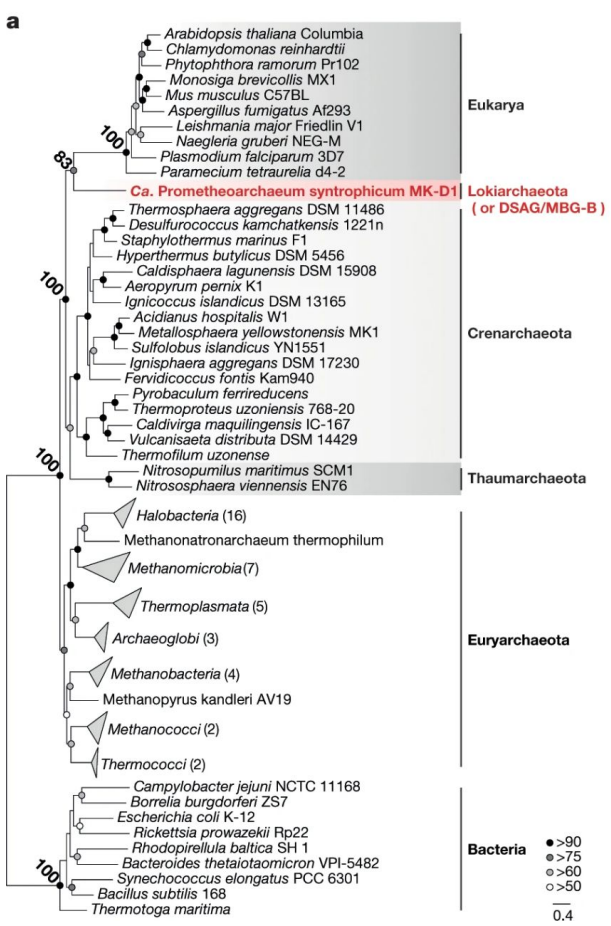
Lokiarchaeota belongs to a larger group called **Asgard archaea** (named after Norway mythology), which includes other groups like Thorarchaeota, Odinarchaeota, and Heimdallarchaeota. **Lokiarchaeota** is a group of archaea that has played a pivotal role in our understanding of eukaryotic evolution. Discovered in sediment samples from the Loki's Castle hydrothermal vent system in the Arctic Ocean.

These organisms have unique genetic and cellular features that appear to **bridge the gap** between prokaryotes (specifically archaea) and eukaryotes, suggesting they might be closely related to the last common ancestor of eukaryotes.



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The Prokaryote–Eukaryote Interface

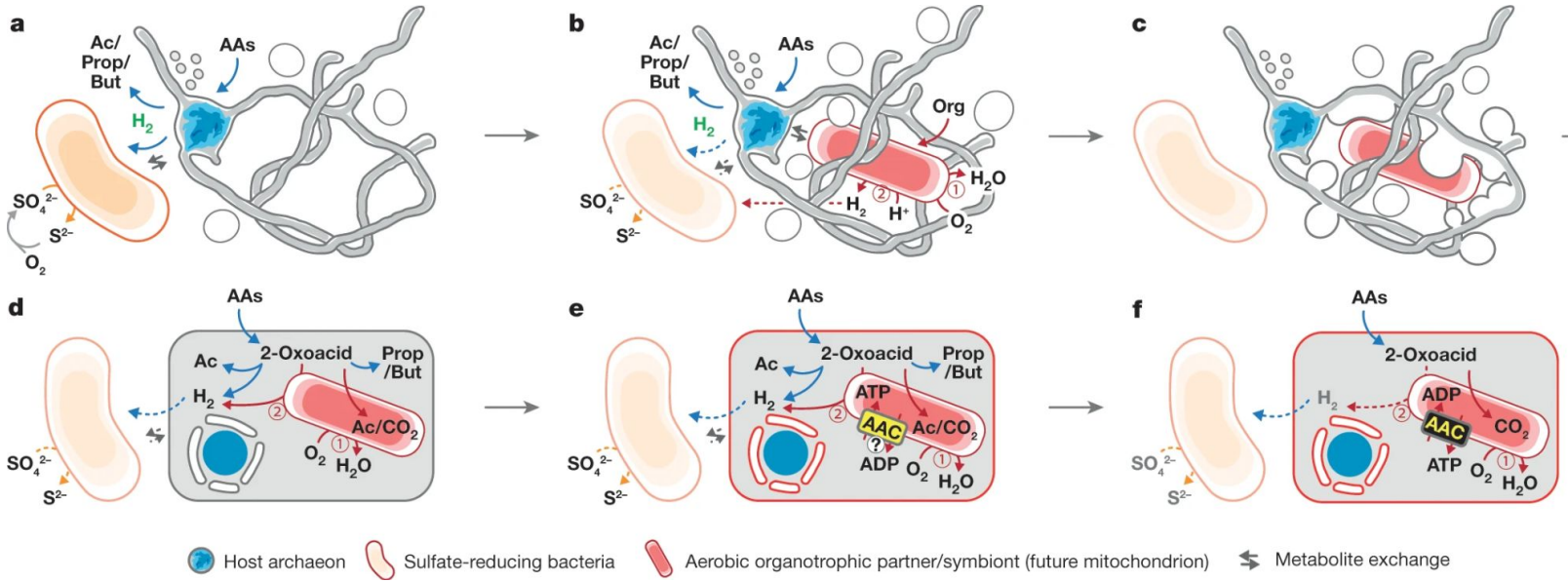


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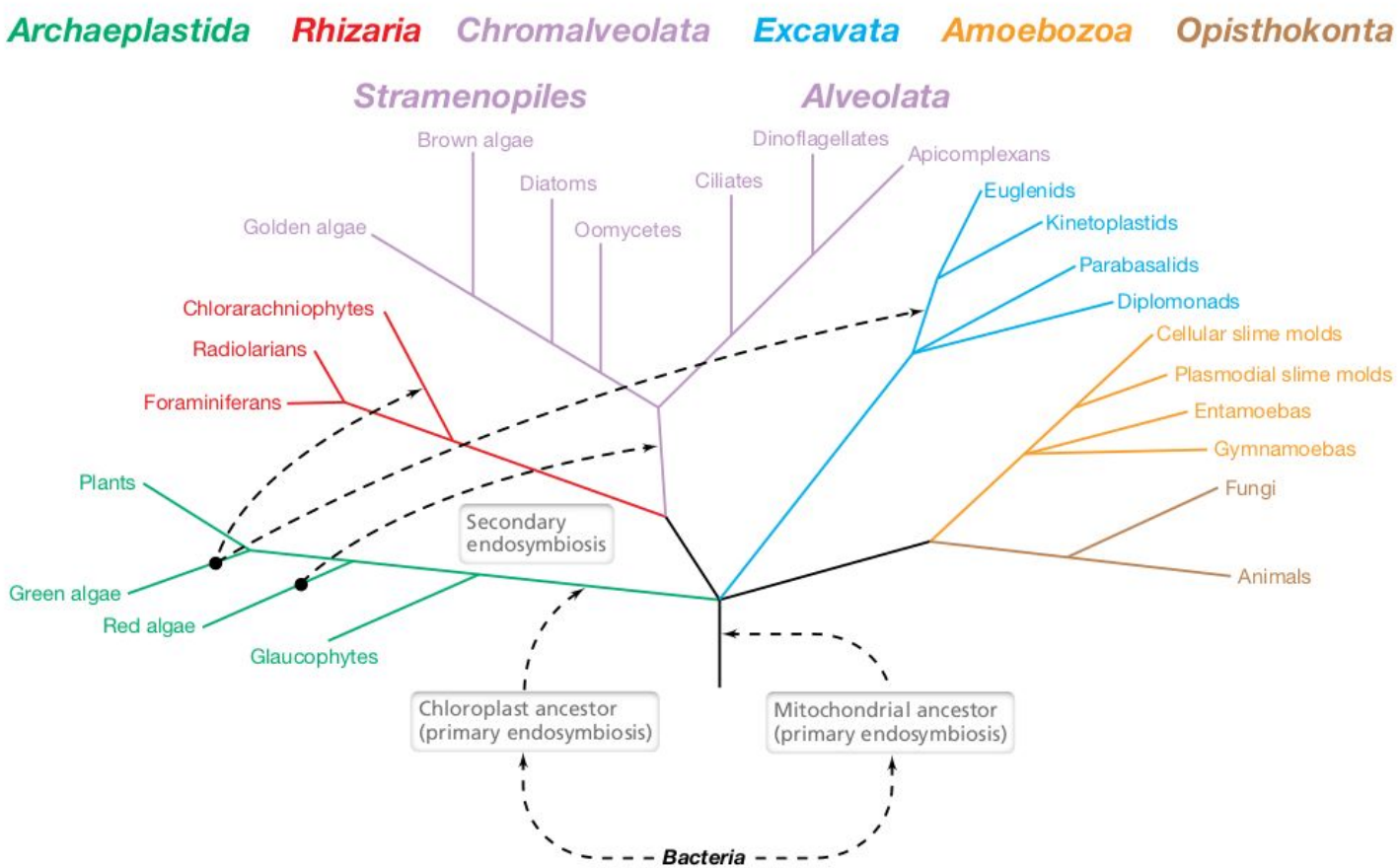
Scientists hypothesize that a simple, oxygen-intolerant Asgard archaeon evolved into the complex ancestor of all eukaryotes by using **outer cell extensions** to engulf an energy-providing, oxygen-consuming bacterial partner, eventually forming mitochondria and internal membranes.



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Phylogenetic Tree Of Eukarya



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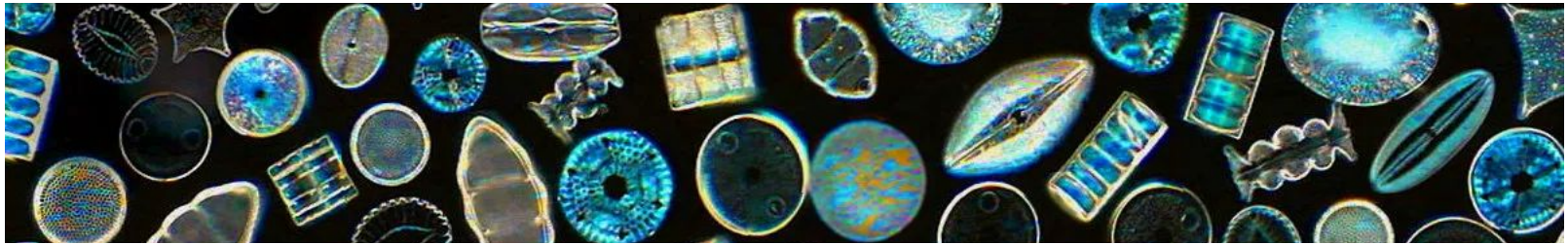
Protists

Protists have fascinated microbiologists since their discovery nearly 350 years ago, representing one of the major groups of eukaryotic microorganisms

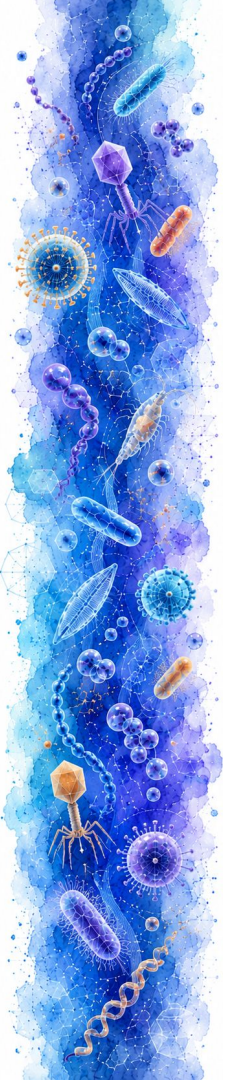
Unicellular, microscopic eukaryotic species with an incredible range of sizes, forms, and functions and, despite their generally diminutive size, constitute much of the genetic diversity within the domain Eukarya

Complex taxonomy with numerous **paraphyletic** groups, common in all eukaryotic lineages except plants, fungi, and animals

In marine ecosystems play fundamental play critical roles in the functioning of food webs and the cycling of carbon and nutrients, being primary producers, consumers, decomposers, and trophic links in aquatic food webs



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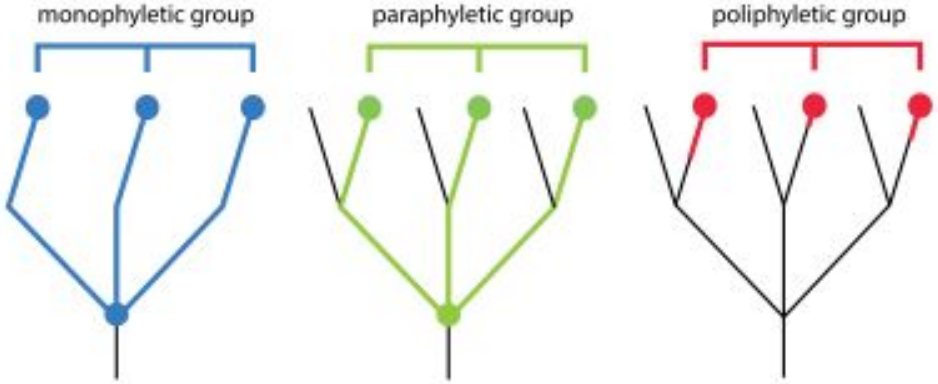
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Taxonomic Zoom In

In evolutionary biology, terms like monophyletic, paraphyletic, and polyphyletic are used to describe the relationships between organisms in phylogenetic trees.



A monophyletic group, or clade, includes an ancestor and all of its descendants. This type of group represents a single, complete lineage, making it the most natural and accurate way to describe evolutionary relationships. EX: Cyanobacteria (all descendants from a single cyanobacterial ancestor)

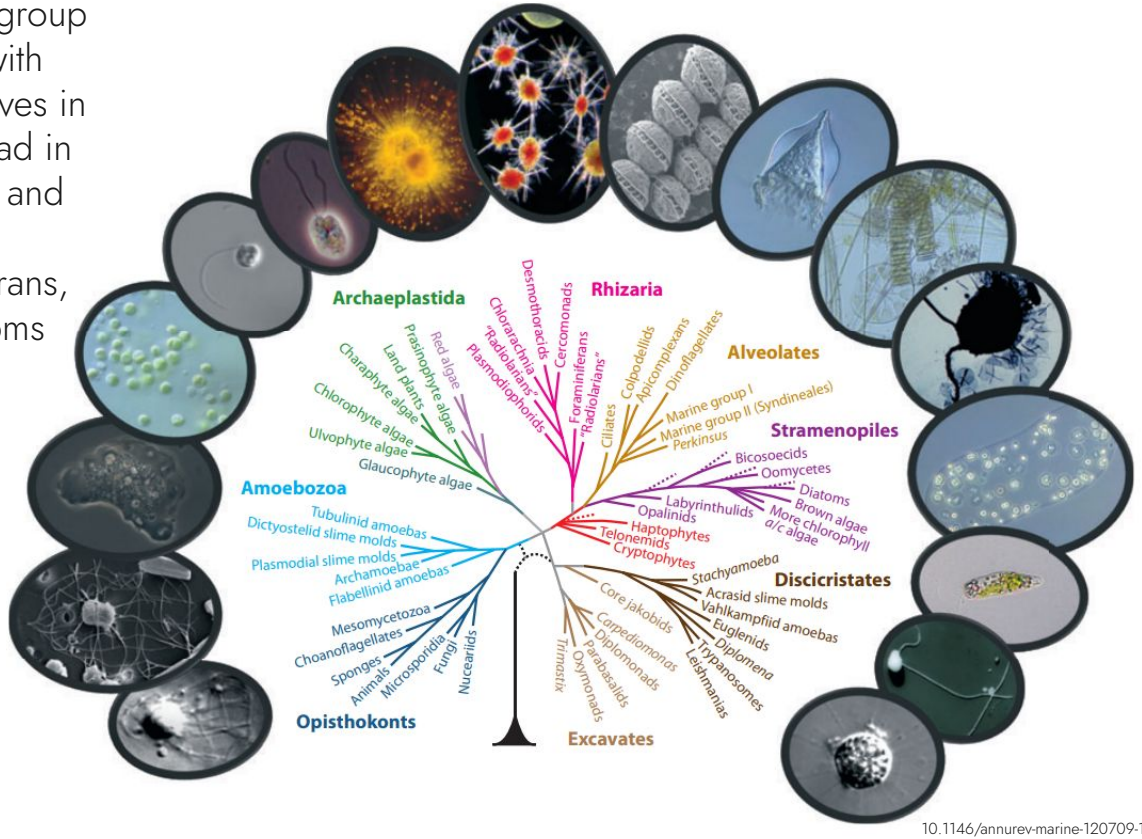
A paraphyletic group includes an ancestor and some, but not all, of its descendants. This grouping leaves out certain lineages that share the common ancestor, often resulting in incomplete or misleading classification. EX: Protists (includes some but excludes fungi, plants, animals)

A polyphyletic group consists of organisms that do not share a recent common ancestor within the group itself. Instead, this grouping is based on convergent traits or similar characteristics that evolved independently, not due to shared ancestry. EX: Anaerobic microbes (methanogens and sulfate-reducing bacteria evolved anaerobically but independently)

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Protists Distribution In The Eukaryotic Phylogenetic Tree

Protists occur in every supergroup within the domain Eukarya, with common marine representatives in Choanoflagellates, widespread in Amoebozoa, in Chlorophyte and Prasinophyte algae, in Cercozomonads, Foraminiferans, Dinoflagellate, Ciliates, Diatoms and so on.



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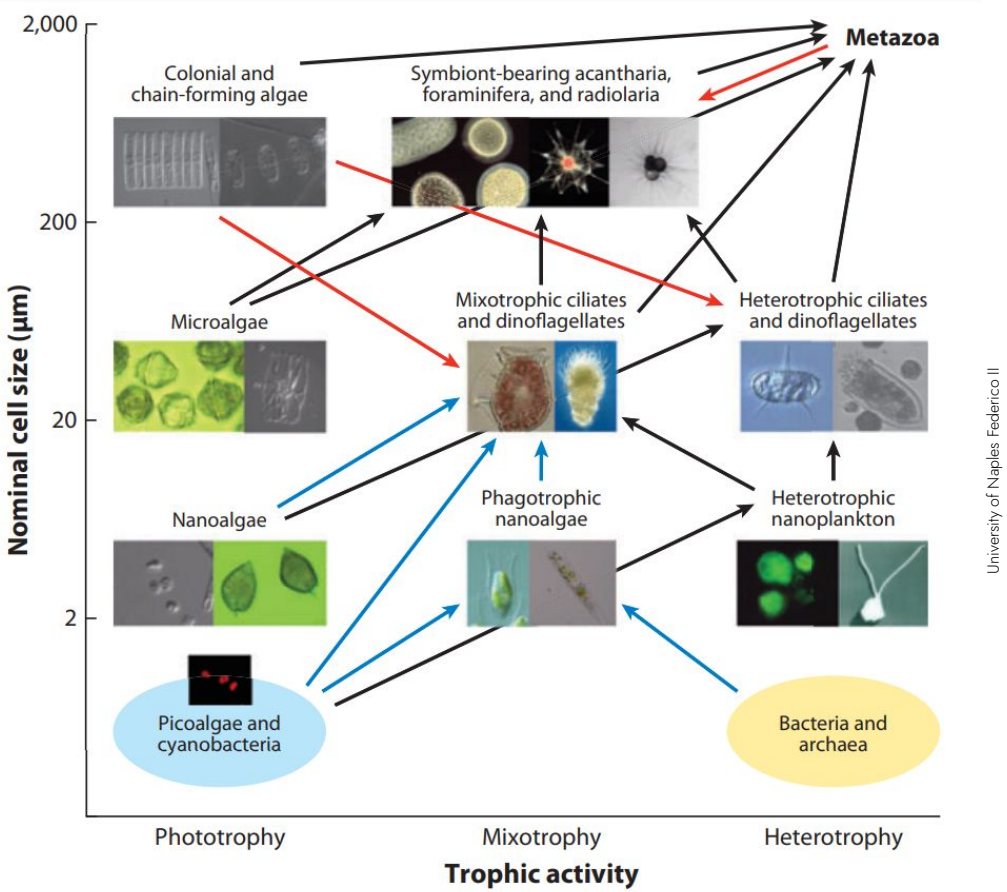
10.1146/annurev-marine-120709-142802

Protists In Marine Food Webs

Protists play pivotal ecological roles as primary producers and consumers at the base of marine food webs.

In food web models they are generally grouped by size and by the nutritional modes of phototrophy, heterotrophy, and mixotrophy. Arrows depict trophic interactions, indicating the direction of carbon flow during the consumption of prey by heterotrophic species (black), mixotrophic species (blue), or the direction of carbon flow when larger prey are consumed by smaller predators (red).

Still little is known about protists biodiversity and ecological roles, and this knowledge gap is defined "protistean gap".

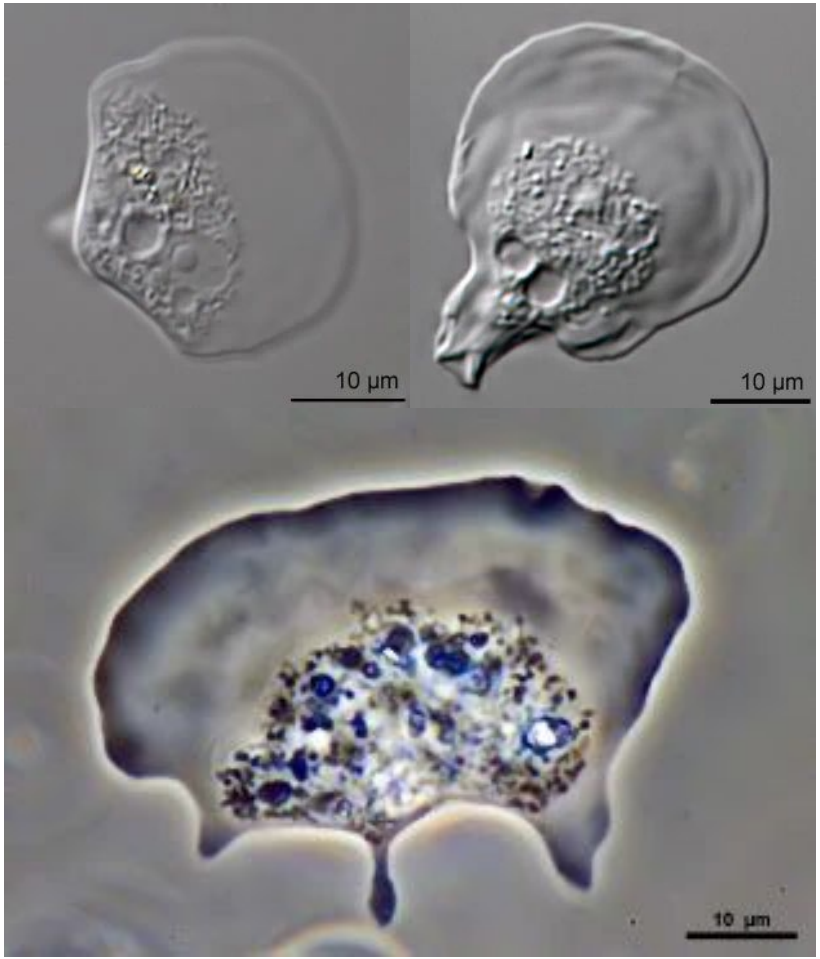


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Amoebozoa: Vannellidae

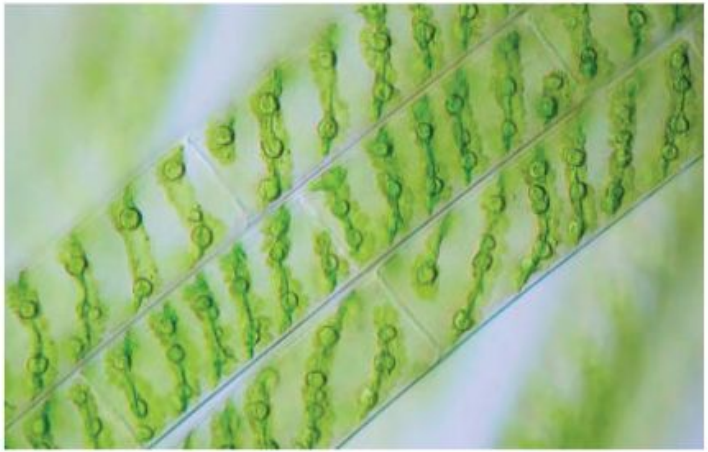
The Vannellidae are a family of Amoebozoa, which are found in soil, fresh- and seawater. They generally have flattened, fan-shaped or spatulate shapes with a large frontal hyaline area, an optically transparent region of cytoplasm. Vannella *spp*., particularly ***Vannella platypodia***, are more representative of the typical marine amoebae. These species thrive in coastal waters and sediments and are frequently isolated from marine environments. They are **heterotrophic bacterivorous** and feed on various microorganisms, including bacteria and small algae. This predatory role helps regulate bacterial populations, maintaining balance in microbial communities and contributing to the recycling of organic and inorganic nutrients, like carbon, nitrogen and phosphorus.



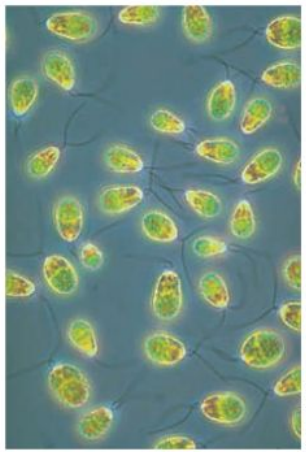
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Archaeplastida: Chlorophyta

Chlorophyta is a taxon of green algae, aquatic photoautotrophic eukaryotic organisms. They have chloroplasts containing chlorophylls a and b, which give them their characteristic green color, but they lack phycobiliproteins and so do not develop the red colors of red algae. Species of Chlorophyta are common inhabitants of marine, freshwater and terrestrial environments. Several species have adapted to specialised and extreme environments, such as deserts, arctic environments, hypersaline habitats, marine deep waters, deep-sea hydrothermal vents and habitats that experiences extreme changes in temperature, light and salinity. Several species of Chlorophyta live in symbiosis with a diverse range of eukaryotes, including ciliates, forams, cnidarians and molluscs. Some species of Chlorophyta are heterotrophic, either free-living or parasitic. Others are mixotrophic bacterivores through phagocytosis.



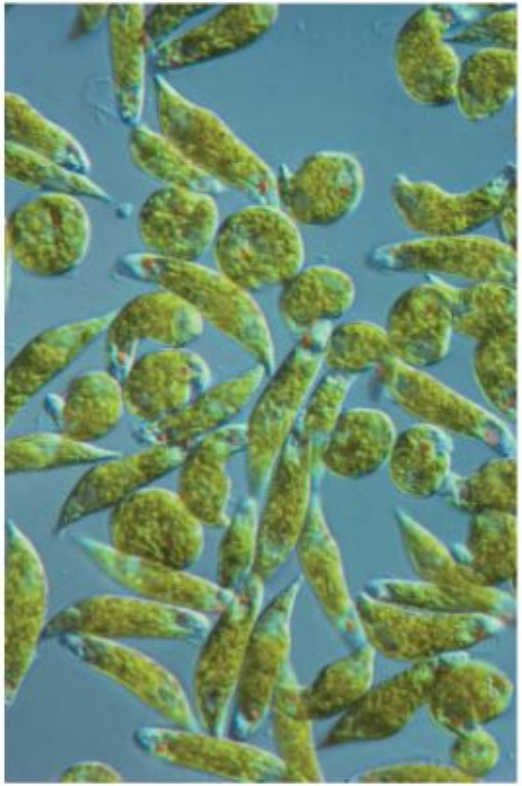




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Excavata: Euglenids

A well-studied group of Excavata are the Euglenids, motile microbial non-pathogenic eukaryotes. Euglenids provided one of the earliest challenges to traditional "zoo-" and "phyto-" based old classification, since their nutrition may be phototrophic or heterotrophic (phagotrophic), or combine features of both (mixotrophy). The photosynthetic euglenids, the class Euglenophyceae, acquired a green plastid through secondary endosymbiosis. They mostly occur in freshwater habitats, but a few marine species are also known (*Eutreptia viridis*)

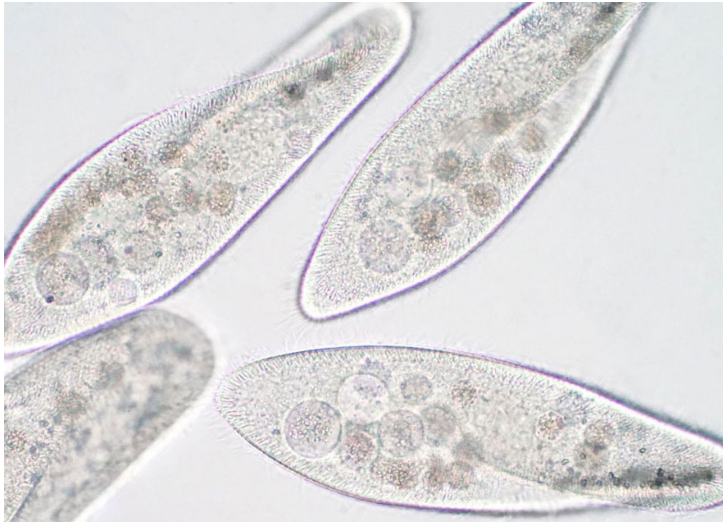


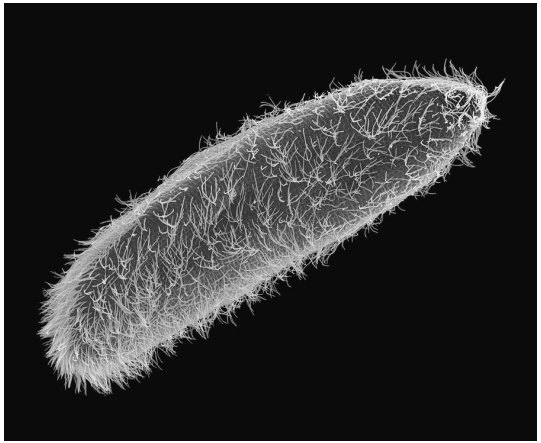


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Alveolata: Ciliates

The Ciliates are a group in the Alveolata characterized by the presence of hair-like organelles called cilia. Cilia occur in all members of the group and are variously used for swimming, crawling, attachment, feeding, and sensation. Ciliates are predominantly heterotrophic, obtaining their nutrients by ingesting organic matter, including bacteria, algae, and small protozoa. They are highly abundant in the oceans, both in open waters and in bottom sediments, and more is known about them than other groups because they have been studied microscopically for many years, due to their larger size (15-80 µm, with some up to 200 µm). Interest in ciliates has increased in recent years because of growing recognition of their essential role in the microbial loop of ocean food webs, being both prayed and predator. At least 8000 species of ciliates are known, many of which are marine, with the best-known and most widely distributed belonging to the genus Paramecium.





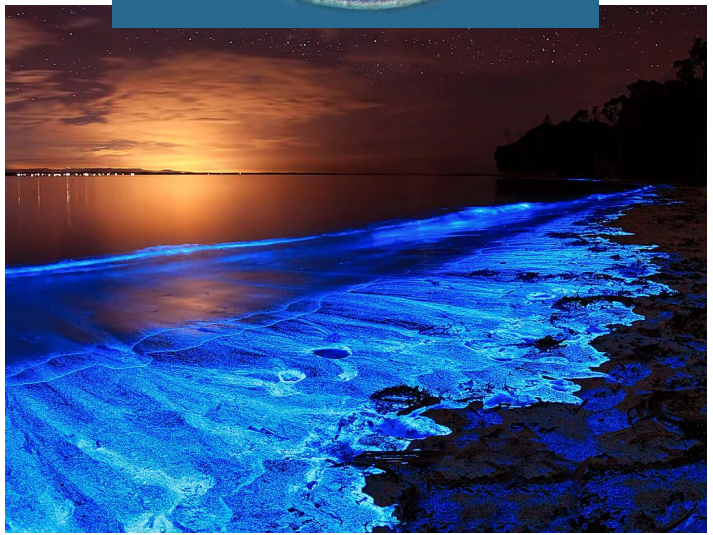
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Alveolata: Dinoflagellates

Dinoflagellates are a diverse group of marine and freshwater phototrophic alveolates. Dinoflagellates have two flagella of different lengths and with different points of insertion into the cell, transverse and longitudinal. Flagella encircling the cell impart spinning movements that give dinoflagellates their name ("δῖνος" in Greek means "whirling").

Some dinoflagellates are free-living, whereas others live a symbiotic existence with animals that form coral reefs, obtaining a sheltered and protected habitat in exchange for supplying phototrophically fixed carbon as a food source for the reef. A number of free-living species are capable of bioluminescence and emit light when disturbed at night; this bioluminescence results in a "sparkling" effect that can often be observed in coastal seas and bioluminescent bays.



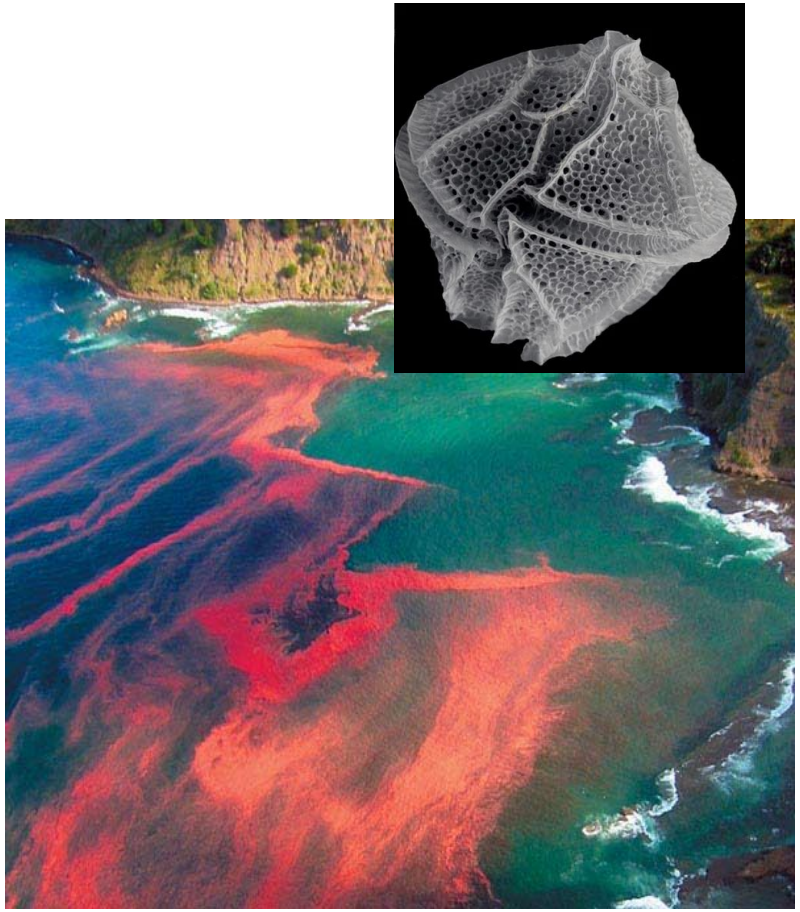


Noctiluca scintillans in Arabian Sea, India

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Alveolata: Dinoflagellates

Several species of dinoflagellates are **toxic**.
For example, dense suspensions of Gonyaulax spinifera cells, called "red tides" because of the red-colored pigments of this organism, can form in warm and typically polluted coastal waters. Such blooms are often associated with fish kills and poisoning in humans following consumption of mussels that have accumulated Gonyaulax through filter feeding. Toxicity results from a neurotoxin (saxitoxin) that can cause a condition called "paralytic shellfish poisoning" in humans and some marine animals. Symptoms include numbness of the lips, dizziness, and difficulty breathing; in severe cases, death can result from respiratory failure.

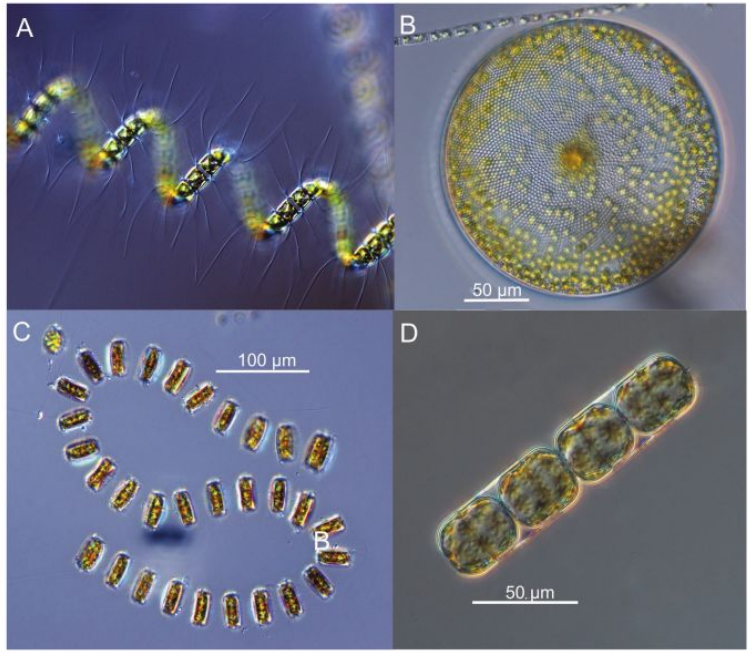


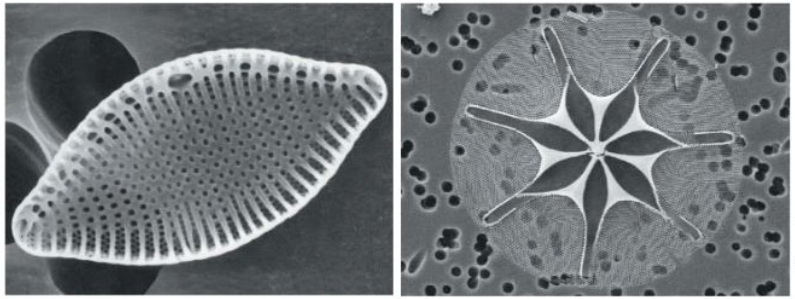
Gonyaulax spinifera in Sabah waters, Malaysia

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Stramenopiles: Diatoms

The diatoms are one of most diverse groups of the protists, with in excess of 200 genera and about 105 current species. They are unicellular, phototrophic, microbial eukaryotes, and are major components of the free-living phytoplankton microbial community in marine environment, but many attach to surfaces such as rocks, marine plants, mollusks, crustaceans, and larger animals. Diatoms characteristically produce a porous cell wall made of silica, which protects the cell against predation, exhibits widely different shapes in different species and can be highly ornate. Diatoms are largely responsible for the spring bloom along the continental shelf in temperate waters. A key factor determining the size of blooms is the availability of nutrients, especially nitrogen and phosphorus, but the concentration of silica is also a critical limiting factor for diatoms, and most will not grow at silica concentrations below about 2 µM.

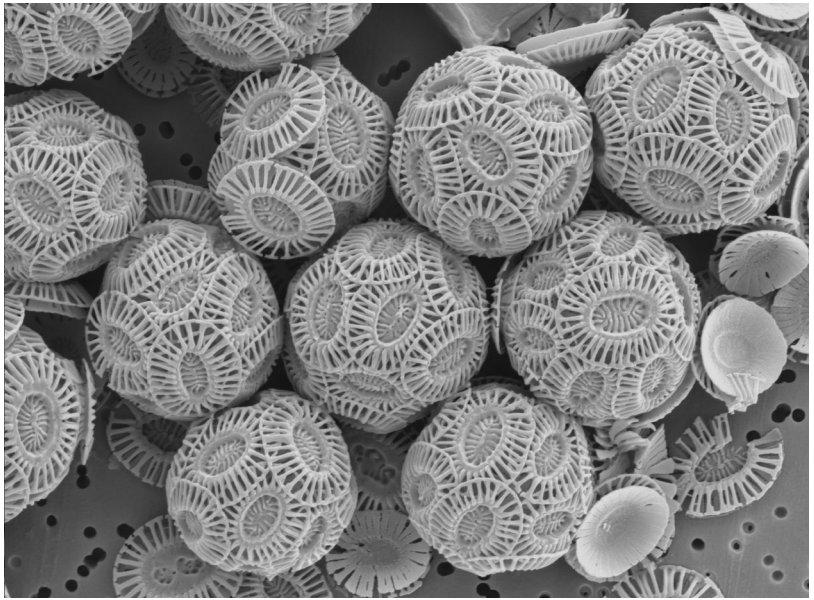




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Diaphoretickes: Haptophyte

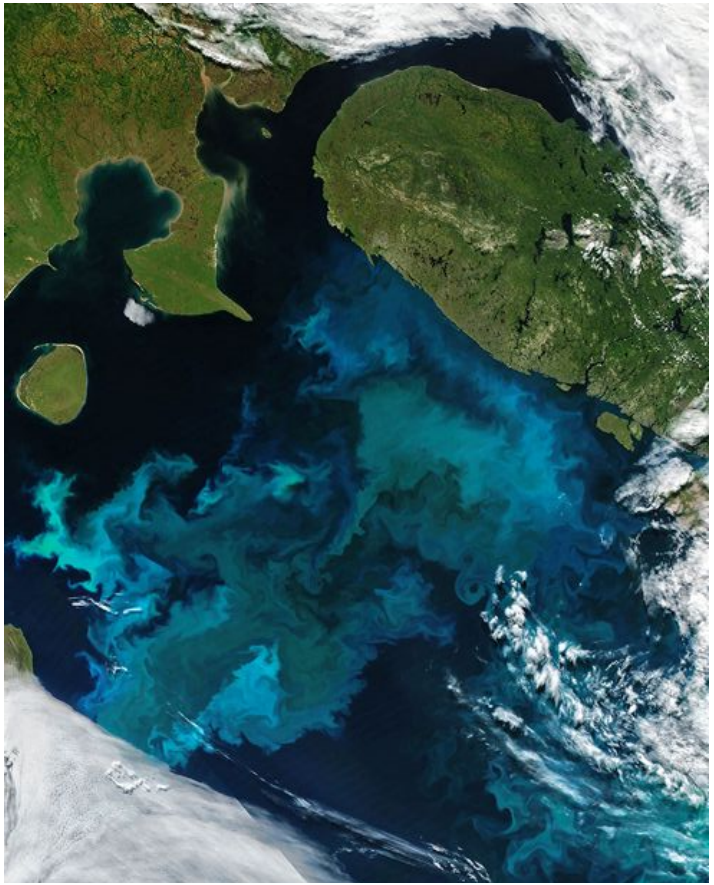
The Haptophytes, also known as prymnesiophytes, are very important members of the phytoplankton, phylogenetically included in the Clade of Diaphoretickes, but with rank classification still under debate. There are about 500 living species in 50 genera, among which most species are marine and occurring worldwide, with the greatest diversity and abundance in tropical waters.
The best-known members of the haptophytes are the **Coccolithophores**, unicellular photoautotrophic organisms which have been one of the most abundant forms of phytoplankton in the oceans for millions of years. Coccolithophores cover themselves with a calcium carbonate shell called a coccosphere, probably as protection against microzooplankton predation. The most abundant example of coccolithophores is ***Emiliania huxleyi***.



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Diaphoretickes: Haptophyte

Each year, massive blooms of *Emiliania huxleyi* occur in the upper photic layers of temperate and sub-temperate waters. They are especially notable in the coastal waters of northern Europe and Scandinavia following spring diatom blooms, where they are highly visible as milky- white areas in satellite images, because of the reflective properties of coccoliths. Under favorable conditions, individual blooms may cover 105 km^2 and contain more than 10^{21} cells, accounting for almost 90% of all the phytoplankton in the upper water columns. The reflective "signature" of the coccoliths is unique and allows precise measurement of the fate of the bloom from satellite images. On a global scale, *E. huxleyi* is also present at relatively constant, lower levels in many oligotrophic oceans, where they also contribute significantly to primary productivity.



Emiliania huxleyi in Barents Sea, Russia

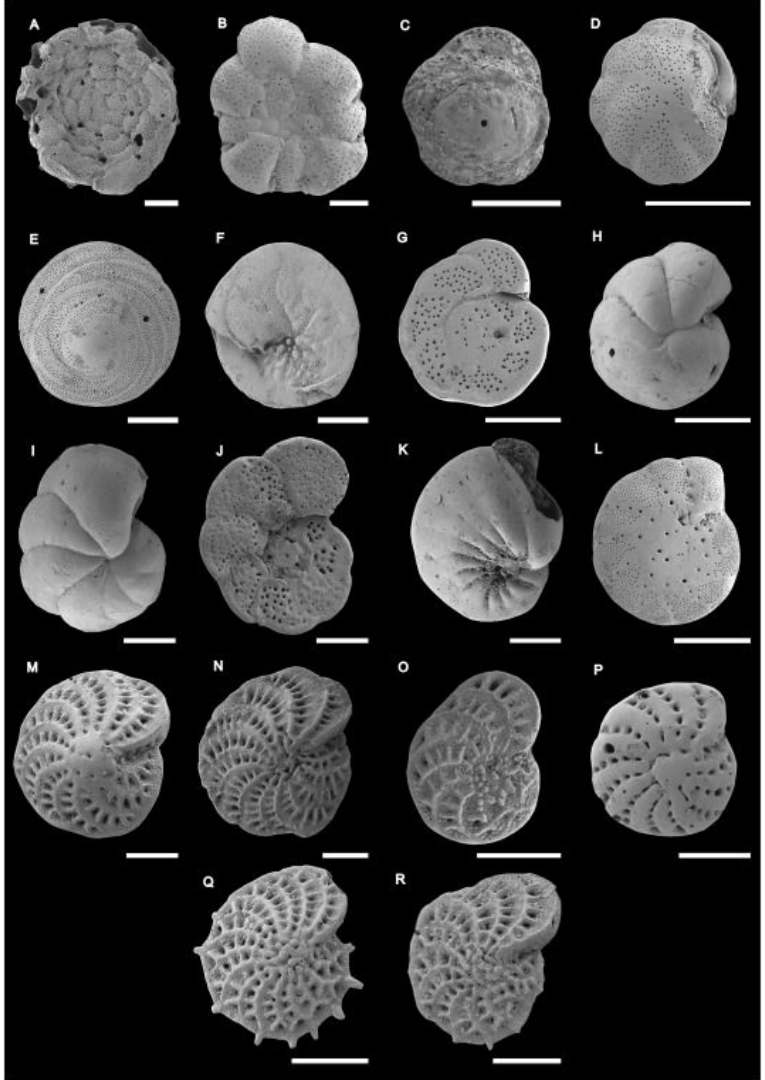
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Rhizaria: Foraminifera

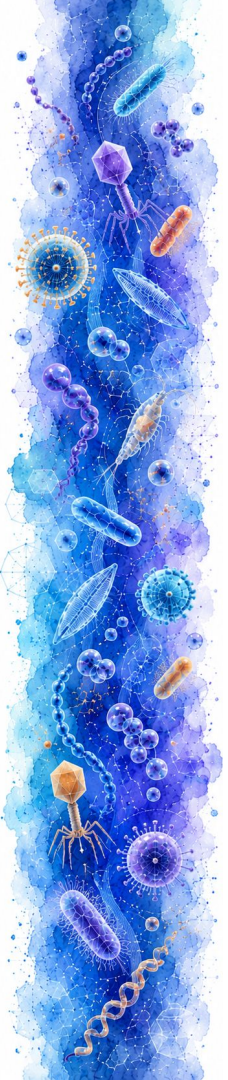
Foraminifera are unicellular members of the class Rhizaria, characterized by streaming granular ectoplasm for catching food and other uses, and commonly by an external shell, called "**test**", of diverse forms and materials.

These tests may consist of one of multiple chambers, and may be composed of protein, sediment particles, calcite, aragonite, or silica. Foraminifera have significant application in the field of biostratigraphy. Due to their small size and hard shells, foraminifera may be preserved in great abundance and with high quality of preservation.

Calcareous fossil foraminifera are very useful in paleoclimatology and paleoceanography. They can be used, as a climate proxy, to reconstruct past climate by examining the stable isotope ratios and trace element content of the tests.



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Rhizaria: Foraminifera

The incorporation of trace elements and stable isotopes into foraminiferal calcite varies systematically with environmental parameters such as temperature, salinity, carbonate chemistry and oxygenation. Measurements of Mg/Ca ratios in shell walls serve as a paleothermometer, whilst oxygen isotope ratios ($\delta^{18}O$) integrate sea surface temperature and global ice volume.

Mg/Ca ratios

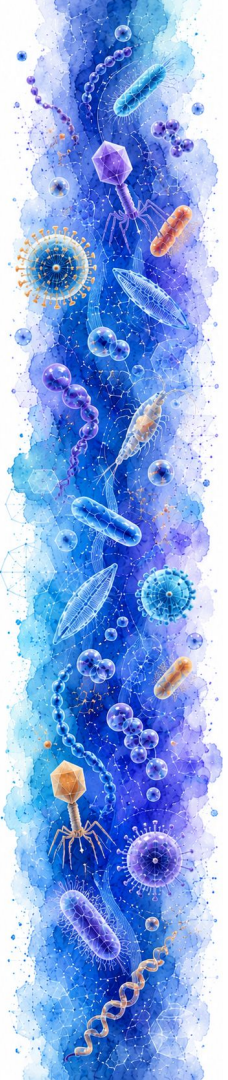
The substitution of Ca by Mg is fundamentally controlled by temperature. As seawater temperature rises, foraminifera incorporate exponentially more Mg into their shells.

Magnesium Mg^{2+} and calcium Ca^{2+} behave similarly in chemical reactions (alkaline earth metals sharing the same valence electron structure). Because of this similarity, Mg^{2+} can act as an "impurity" and substitute for Ca^{2+} within the crystal framework of the calcite $CaCO_3$ as the foraminifera build their shells.

The chemical insertion of Mg into the calcite lattice is an endothermic reaction, meaning it requires heat absorption. As such, warmer water carries higher thermal energy and drive this endothermic reaction, allowing more Mg^{2+} ions to overcome the energy barrier and lodge into the crystal framework.
In cooler water, instead, the reaction has less energy to proceed, resulting in less Mg substitution.

Because of this thermodynamic rule, the Mg/Ca ratio in the shell increases exponentially as growth temperature increases (typically rising by about 6% to 10% per 1 °C increase).

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Rhizaria: Foraminifera

The incorporation of trace elements and stable isotopes into foraminiferal calcite varies systematically with environmental parameters such as temperature, salinity, carbonate chemistry and oxygenation. Measurements of Mg/Ca ratios in shell walls serve as a paleothermometer, whilst oxygen isotope ratios ($\delta^{18}O$) integrate sea surface temperature and global ice volume.

Isotopic fractionation

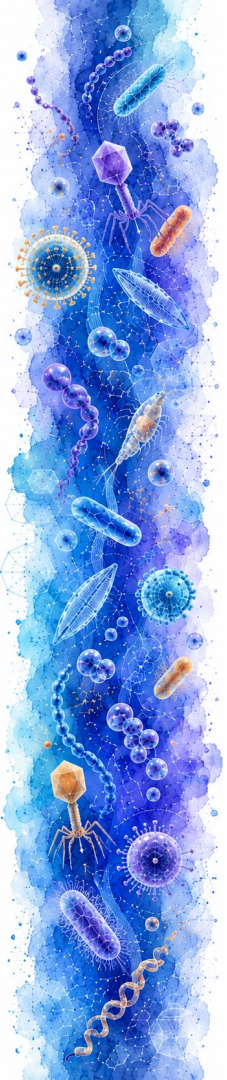
Glacial Periods (Ice Ages): growing glaciers and ice sheets are enriched in ^{16}O, while oceans are enriched in ^{18}O.
Interglacial Periods (Warm Eras): Melting glaciers release ^{16}O back into the ocean, which lowers the ratio of ^{18}O to ^{16}O in the sea.

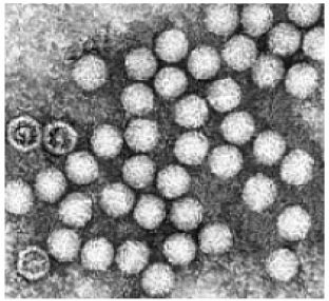
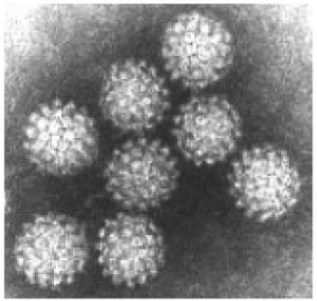
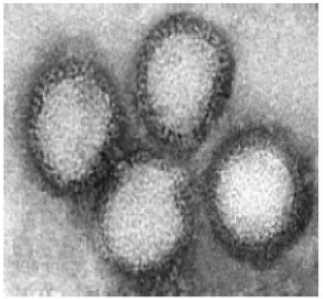
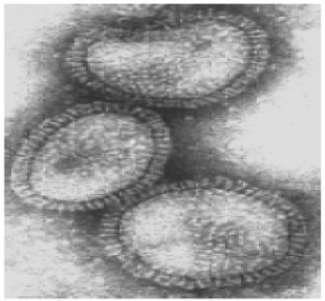
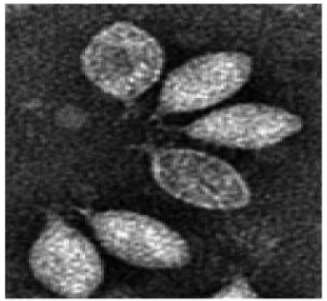
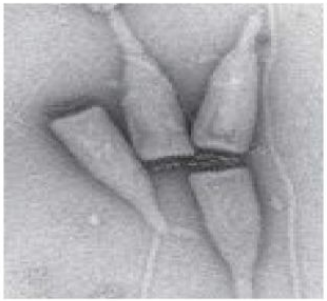
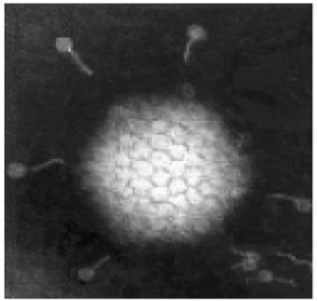
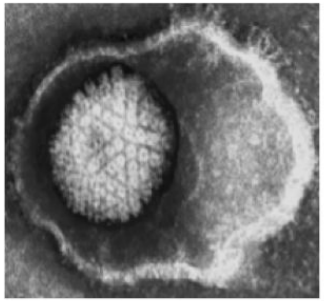
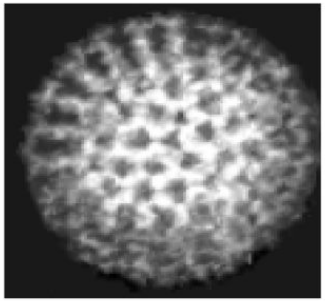
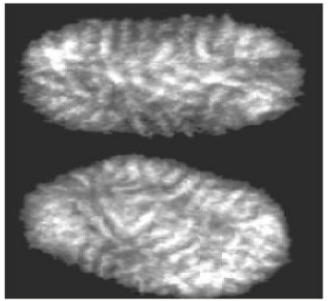
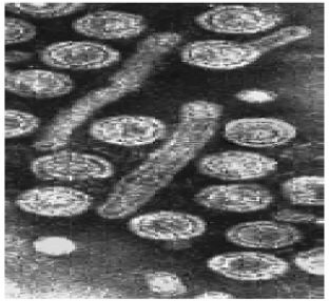
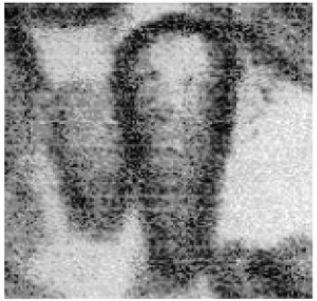
Oxygen incorporation into carbonate is temperature-dependent:
Cold water speed up ^{18}O incorporation into the shells. For each 1 °C drop in ocean temperature = $^{18}O_{CaCO3\ shells}$ increases by ~0.22‰.

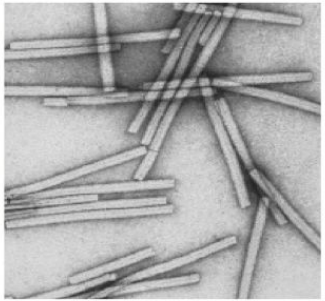
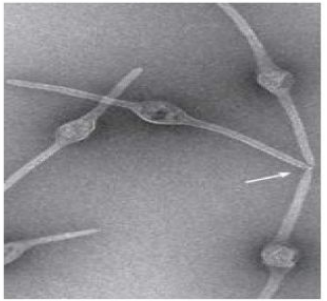
Higher $\delta^{18}O$ → Colder temperatures and larger global ice volume
Lower $\delta^{18}O$ → Warmer temperatures and smaller global ice volume

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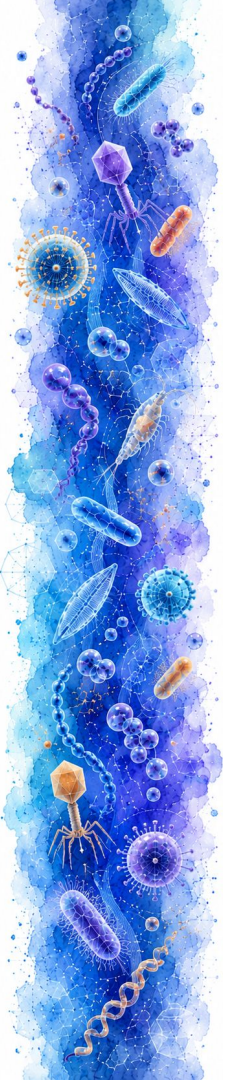
Viruses



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Viruses

.The word virus comes from the Latin, poison (syn. Venenum)

.A virus is a genetic element that can replicate only inside a host cell, and therefore cannot reproduce on its own

.Although viruses are not cells, they nonetheless contain a genome, that encodes those functions needed to replicate, and they have an extracellular form, called “virion”, which allows the virus to travel from one host cell to another

.The virion of any virus includes a protein shell, called the capsid, in which the viral genome is stored

.They infect a wide variety of organisms: both eukaryotes and prokaryotes. A virus that infects bacteria is known as a bacteriophage, often shortened to phage

.Simply stated a virus is a “piece of bad news wrapped in a protein case”

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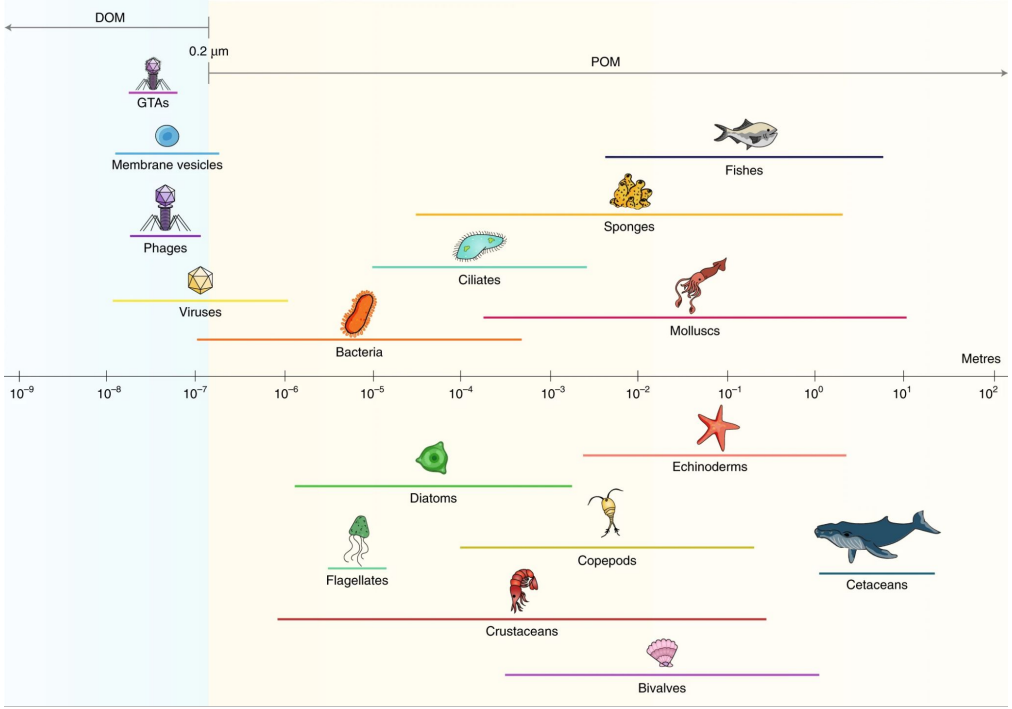
Virus Size

Most viruses range from 0.02 to 0.3 micrometers (μm), or 20 to 300 nanometers (nm), in diameter.

The smallest viruses measure around 0.017 to 0.02 μm, such as the Porcine Circovirus.

Average size viruses measure around 0.03 to 0.1 μm, such as poliovirus, rhinovirus, and influenza.

Giant viruses can reach 0.4 to 1.5 μm in length or diameter, overlapping the size of small bacteria. *Bacillus megaterium* phage G, 160 μm capsid diameter



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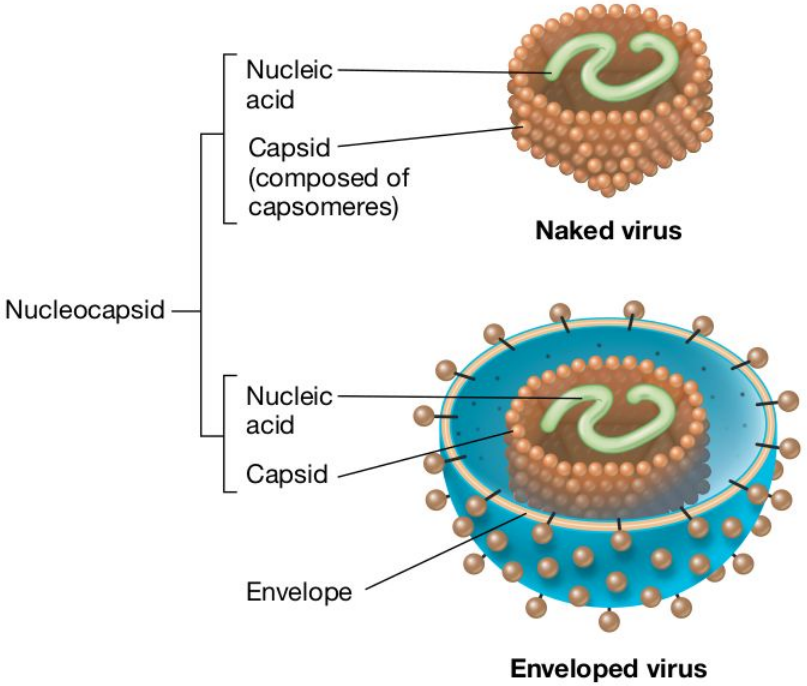
https://doi.org/10.1038/s41564-018-0166-y

Viral Structures And Morphologies

Most bacterial viruses are naked, with no further layers, whereas many animal viruses have an outer layer called the envelope that consists of a phospholipid bilayer (taken from the host cell membrane) and viral proteins.

In enveloped viruses, as in naked viruses, the inner structure of nucleic acid plus capsid protein is called the nucleocapsid.

Viral genomes consist of either DNA or RNA and can be can be either linear or circular, single-stranded or double-stranded.



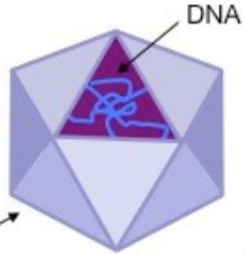
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Viral Structures And Morphologies

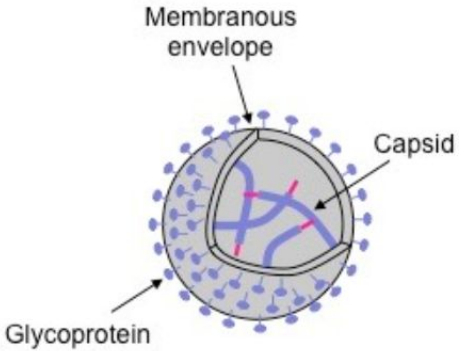
Helical



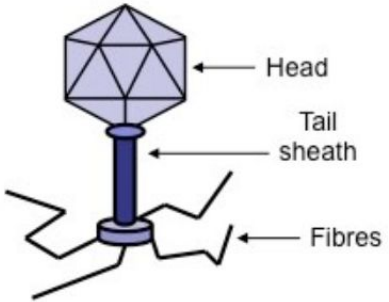
Icosahedral



Spherical enveloped

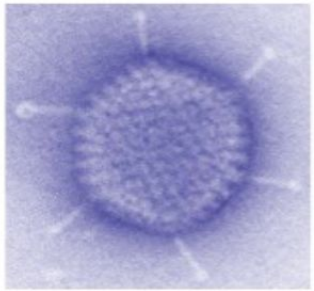


Complex head-and-tail

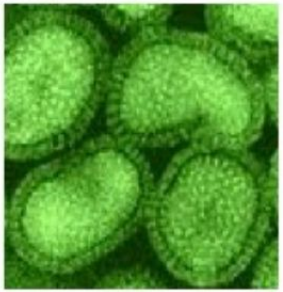




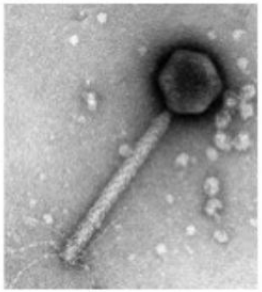
Tobacco Mosaic Virus



Adenovirus



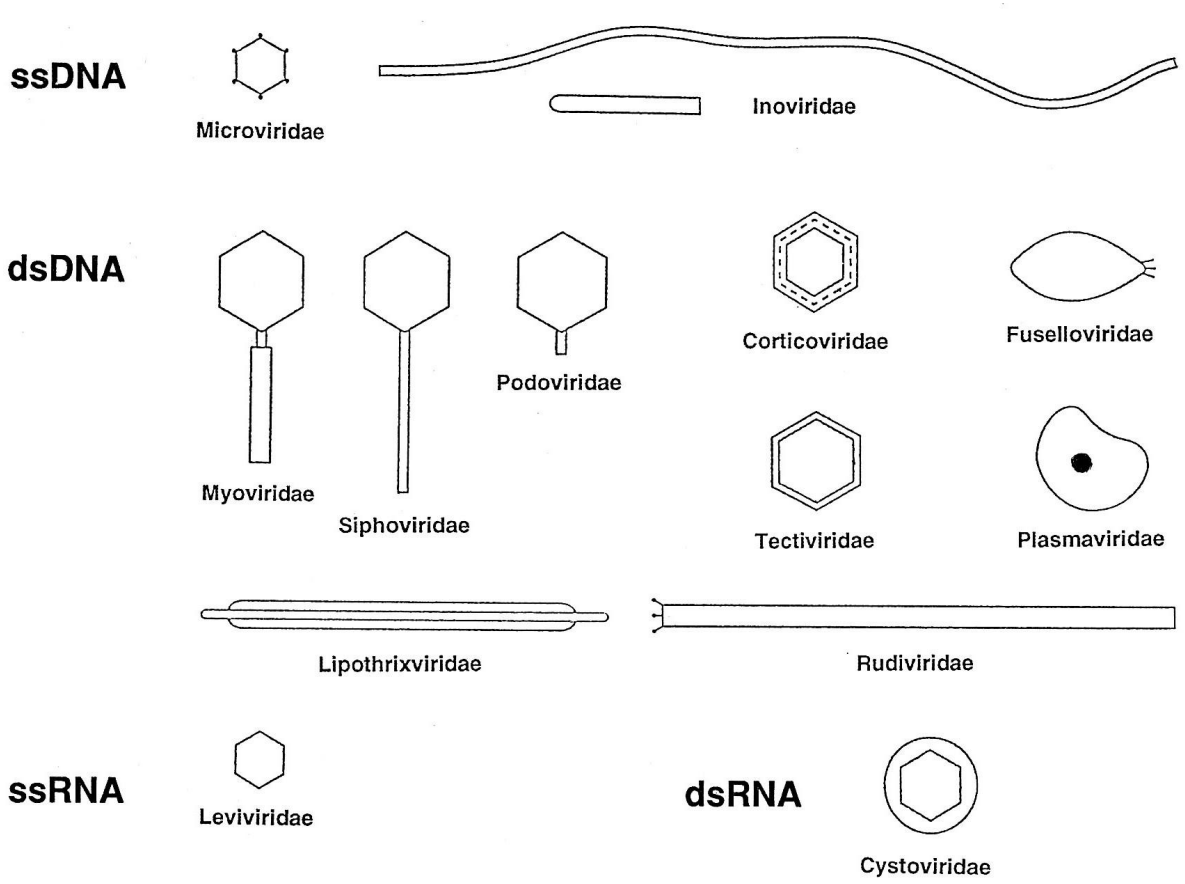
Influenza Virus



Bacteriophage

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Virus Classification



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Virus Classification: Baltimore Classification

All cells contain double-stranded DNA genomes

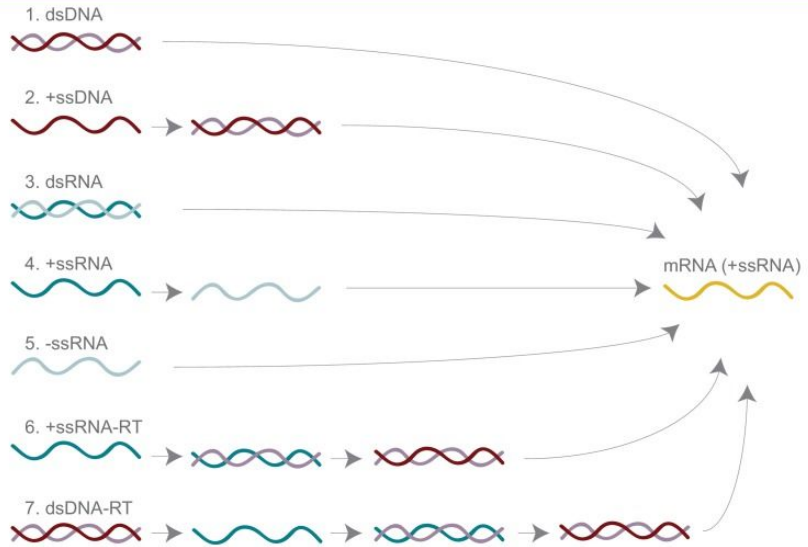
By contrast, viral genomes consist of either DNA or RNA and are further subdivided based on whether the genome is single-stranded or double-stranded

Baltimore Classification:

Group I: Double-stranded DNA
Group II: Single-stranded DNA

Group III: Double-stranded RNA
Group IV: Single-stranded RNA (+)
Group V: Single-stranded RNA (-)

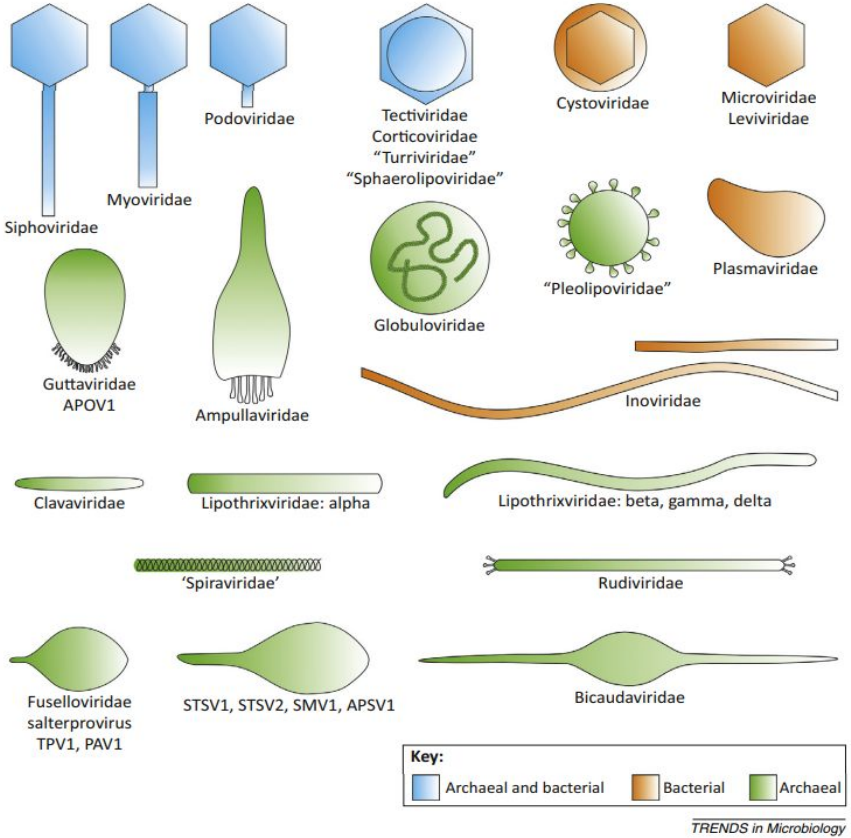
Group VI: Single-stranded RNA + Rev. Transcriptase
Group VII: Double-stranded DNA + Rev. Transcriptase



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https://doi.org/10.1006/rwvi.1999.0277

Viral Classification: Host-Domain Classification



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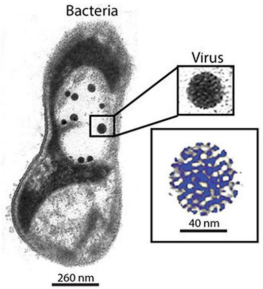
10.1016/j.tim.2014.02.007

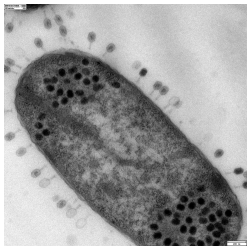
Virus Life Cycle

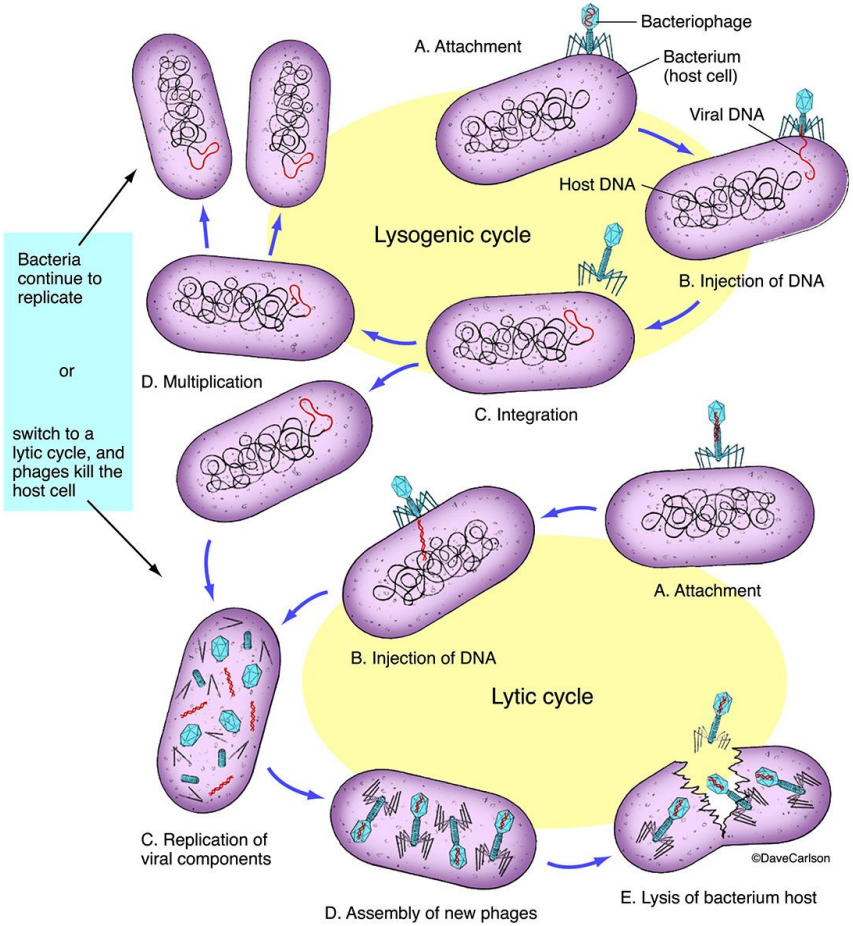
Lytic pathway.
The phage infects and lyse the host cell immediately after reaching burst size

Lysogenic pathway.
The phage becomes dormant (i.e. integrated) into the host genome until induced. Integration can become permanent

Chronic pathway.
The virus remains inside the host cell releasing viral particle without lysing the cell



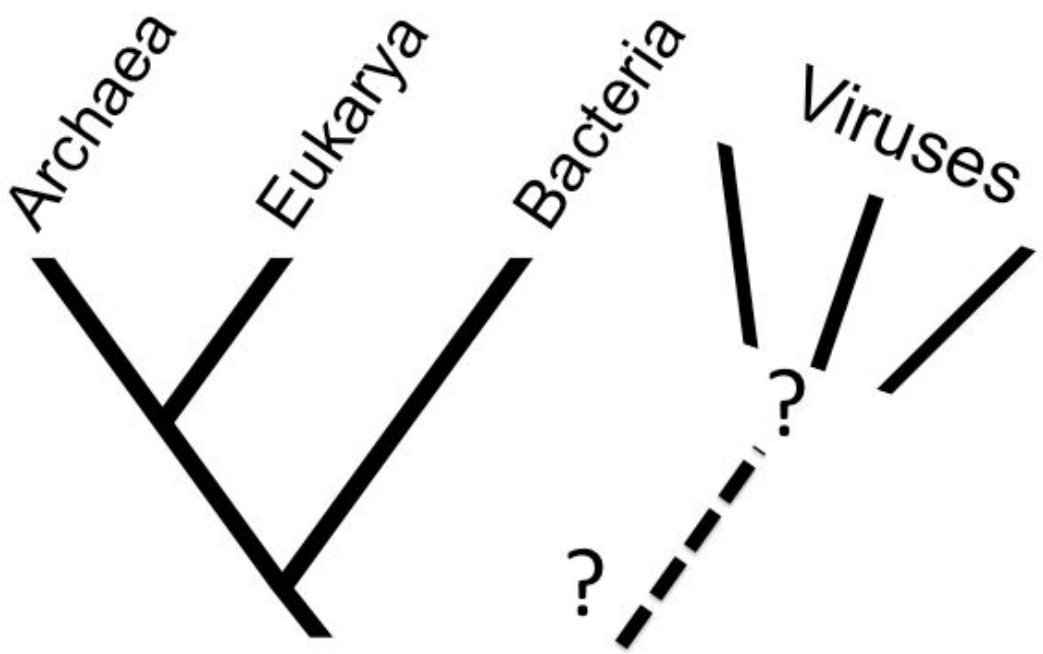




DOI: 10.1590/S2179-975X3720

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Virus And Tree Of Life



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Virus Detection

Vol. 14: 113–118, 1998 | AQUATIC MICROBIAL ECOLOGY Aquat Microb Ecol | Published February 13

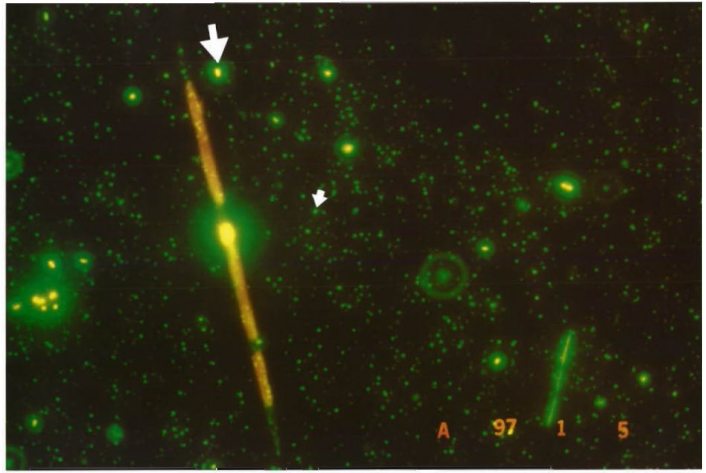
Use of SYBR Green I for rapid epifluorescence counts of marine viruses and bacteria

Rachel T. Noble*, Jed A. Fuhrman

University of Southern California, Department of Biological Sciences, AHF 107, University Park, Los Angeles, California 90089-0371, USA

ABSTRACT: A new nucleic acid stain, SYBR Green I, can be used for the rapid and accurate determination of viral and bacterial abundances in diverse marine samples. We tested this stain with formalin-preserved samples of coastal water and also from depth profiles (to 800 m) from sites 19 and 190 km offshore, by filtering a few ml onto 0.02 µm pore-size filters and staining for 15 min. Comparison of bacterial counts to those made with acridine orange (AO) and virus counts with those made by transmission electron microscopy (TEM) showed very strong correlations. Bacterial counts with AO and SYBR Green I were indistinguishable and almost perfectly correlated ($r^2 = 0.99$). Virus counts ranged widely, from 0.03 to 15×10^7 virus ml^{-1}. Virus counts by SYBR Green I were on the average higher than those made by TEM, and a SYBR Green I versus TEM plot yielded a regression slope of 1.28. The correlation between the two was very high with an r^2 value of 0.98. The precision of the SYBR Green I method was the same as that for TEM, with coefficients of variation of 2.9%. SYBR Green I stained viruses and bacteria are intensely stained and easy to distinguish from other particles with both older and newer generation epifluorescence microscopes. Detritus is generally not stained, unlike when the alternative dye YoPro I is used, so this approach may be suitable for sediments. SYBR Green I stained samples need no desalting or heating, can be fixed with formalin prior to filtration, the optimal staining time is 15 min (resulting in a total preparation time of less than 25 min), and counts can be easily performed at sea immediately after sampling. This method may facilitate incorporation of viral research into most aquatic microbiology laboratories.

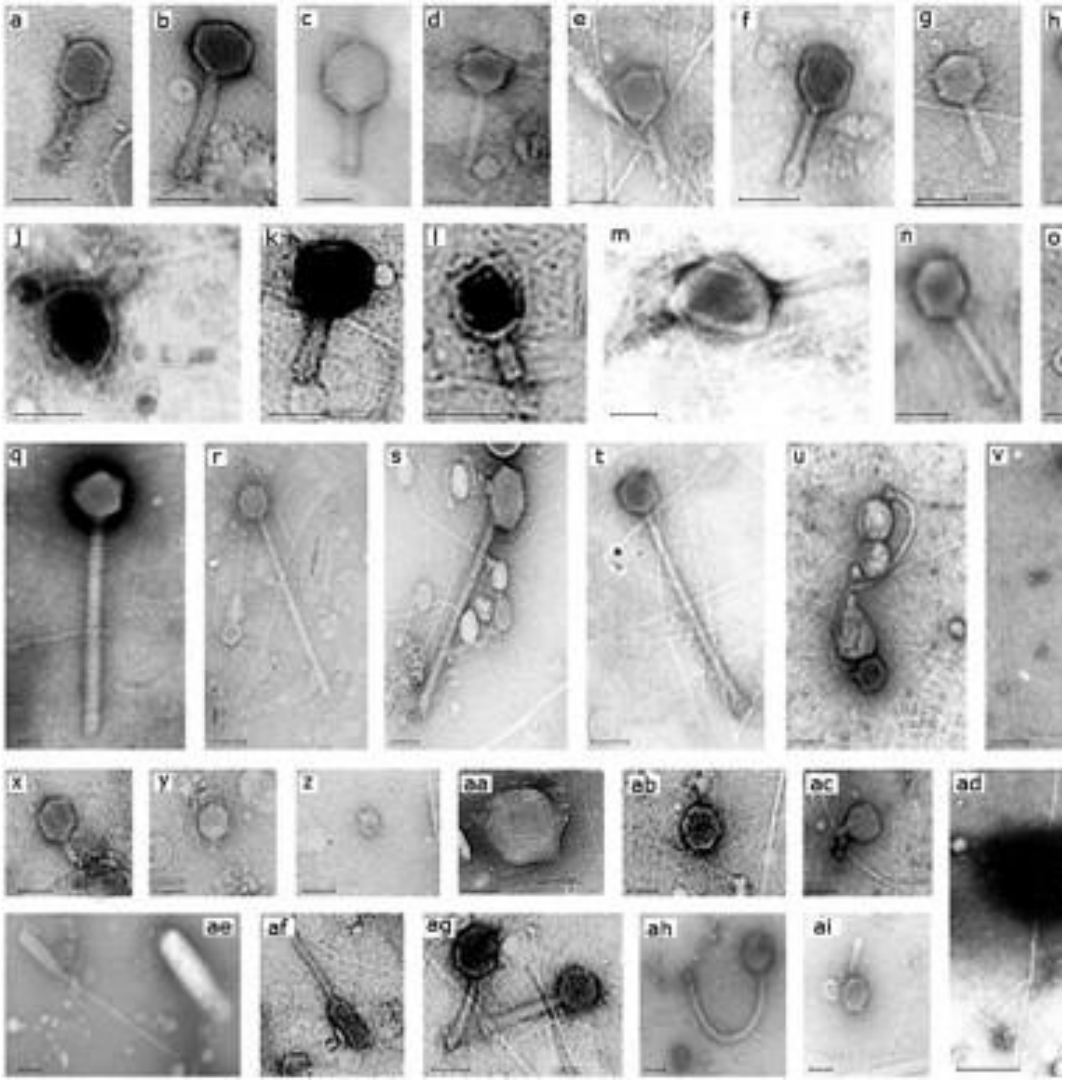
KEY WORDS: Virus · Epifluorescence · SYBR Green I · Marine ecology



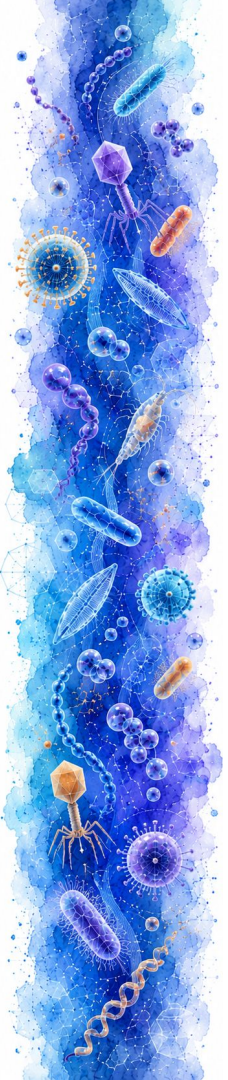
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Virus Detection

Given the lack of conserved genes across the viral groups, for a while the viral diversity was only studied based on their morphologies, using **Scanning (SEM) or Transmission Electron Microscopy (TEM)**, or general genome properties, for example using **Pulse Field Gel Electrophoresis (PFGE)** technique, which is a genotyping technique used for the separation of large DNA molecules (entire genomic DNA).



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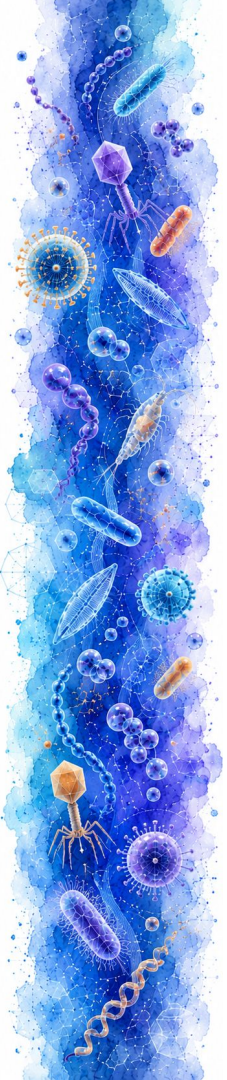
Ten reasons to exclude viruses from the tree of life

1. Viruses are not alive
2. Viruses are polyphyletic
3. There are no ancestral viral lineages
4. Distant hosts do not imply antiquity
5. Viral lineages lack structural continuity
6. Cellular origin of metabolic genes
7. Cellular origin of translation genes
8. Viruses are gene robbers
9. Most HGT occurs from cells to viruses
10. Simplicity does not mean antiquity

Trait	Cells	Viruses
Information content	Yes	Yes
Self-maintenance	Yes	No
Self-replication	Yes	No
Evolution	Yes	By cells
Common ancestry	Yes	No
Structural historical continuity	Yes	No
Genes involved in carbon metabolism	Yes	Cellular origin
Genes involved in energy metabolism	Yes	Cellular origin
Genes involved in protein synthesis	Yes	Cellular origin

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10.1038/nrmicro2108



Reasons to include viruses in the tree of life

"The inability of viruses to self-sustain and self-replicate, their phylogenetic diversity, the origin of their cell-like genes and the instability of their genomes over time were emphasized to strengthen the argument. However, as discussed in this Correspondence, we emphasize that although we might be able to call viruses 'pseudo living entities' or 'molecular parasites' we cannot deprive them of their status as living entities."

"We cannot compare primitive organisms such as viruses with complex living entities such as plants and animals. In fact, because viruses are 'mono-unitary' organisms, the most plausible comparison would be with animal spermatozoa or ova. Can we expect either spermatozoa or ova to survive and replicate in their natural environment without any support? Definitely not."

"Although viruses need host cells for survival and replication, we cannot argue that viruses will not survive in nature if we leave them alone, as hosts themselves are part of the continuum of nature. In fact, even animals and birds are not sustained in nature unless they obtain food from other sources of life, that is, plants or other animals."

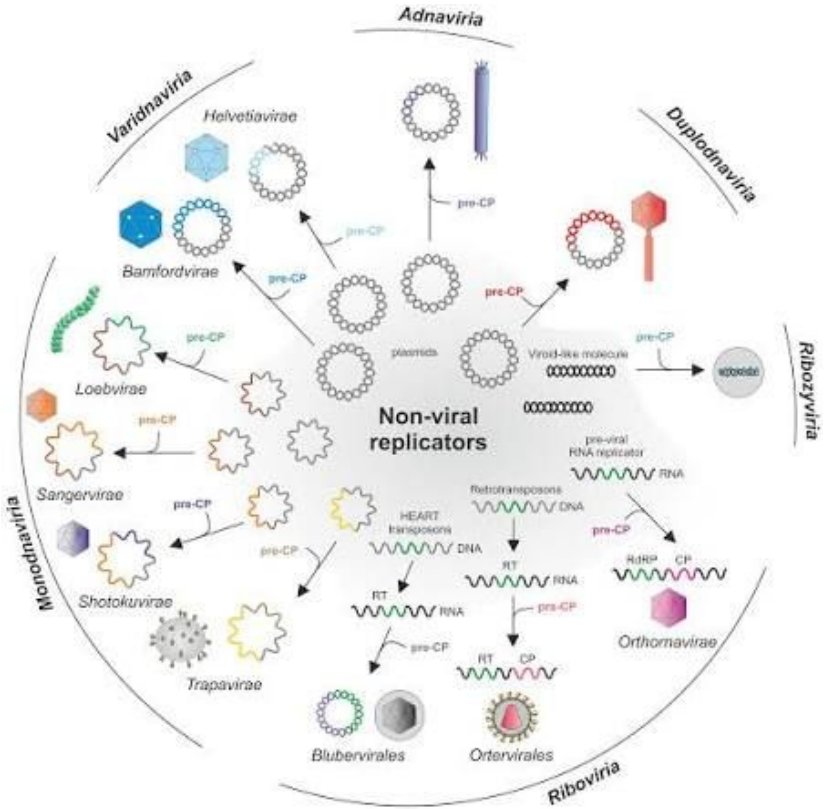
"Pathogen recognition patterns, such as Toll-like receptors, are common features of all living organisms, including plants. In addition, Toll-related proteins of plants recognize viruses. The fact that plants evolved millions of years before animals suggests that ever since the origin of life, living organisms have confronted viruses."

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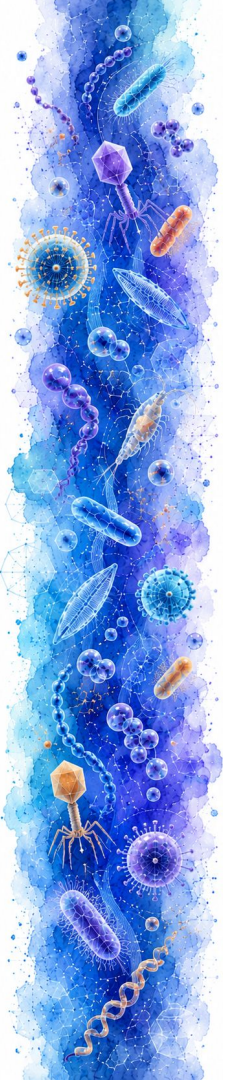
Virus Taxonomy

The **International Committee on Taxonomy of Viruses (ICTV)** is a vital organization responsible for the classification and nomenclature of viruses. Established in 1962, the ICTV aims to create a stable and coherent taxonomic system for viruses, ensuring that the names and classifications are universally accepted and utilized by virologists worldwide. The latest version includes:
10 realms, 11 kingdoms, 22 phyla, 4 subphyla, 50 classes, 105 orders, 14 suborders, 427 families, 232 subfamilies, 4,149 genera, 92 subgenera, **17,554 species**.



https://ictv.global/taxonomy

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Viral Diversity In The Marine Environment

Highest diversity of **unknown genes**, between 40% and 90%, with no matches in databases

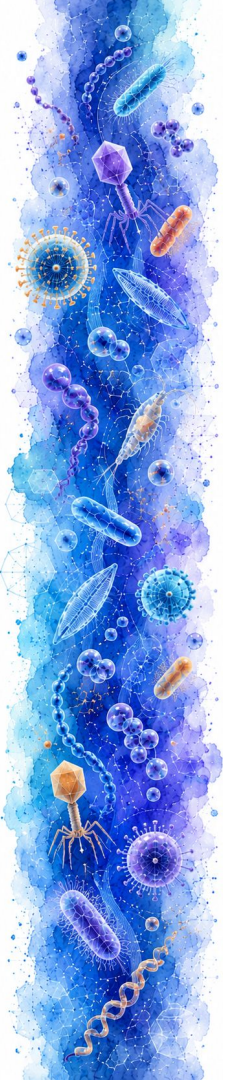
Highest number of **unknown viral families**

Lowest number of "cultured" types available, mostly from model organisms

Huge potential to **influence diversity, evolution and biogeochemistry**

Virus family	Morphology of virion	Size of virion (nm)[a]	Marine host(s)
Double-stranded DNA viruses			
Baculoviridae	Enveloped rods, some with tails	200–450 × 100–400	Crustaceans
Corticoviridae, Tectiviridae,	Icosahedral with spikes	60–75	Bacteria
Herpesviridae	Pleomorphic, icosahedral, enveloped	150–200	Mollusks, fish, corals mammals, turtles
Iridoviridae	Round, icosahedral	190–200	Mollusks, fish
Lipothrixviridae	Thick rod with lipid coat	40 × 400	Archaea
Mimiviridae	Icosahedral with microtubule-like projections	450–650[b]	Protists, corals (?), sponges (?)
Myoviridae	Polygonal head (icosahedral) with contractile tail (helical)	50–110 (head)	Bacteria
Nimaviridae	Enveloped, ovoid with tail-like appendage	120 × 275	Crustaceans
Papovaviridae	Round, icosahedral	40–50	Mollusks
Phycodnaviridae	Icosahedral	130–200	Algae
Podoviridae, Siphoviridae	Icosahedral with noncontractile tail	60 (head)	Bacteria
Single-stranded DNA viruses			
Microviridae	Icosahedral with spikes	25–27	Bacteria
Parvoviridae	Round, icosahedral	20	Crustaceans

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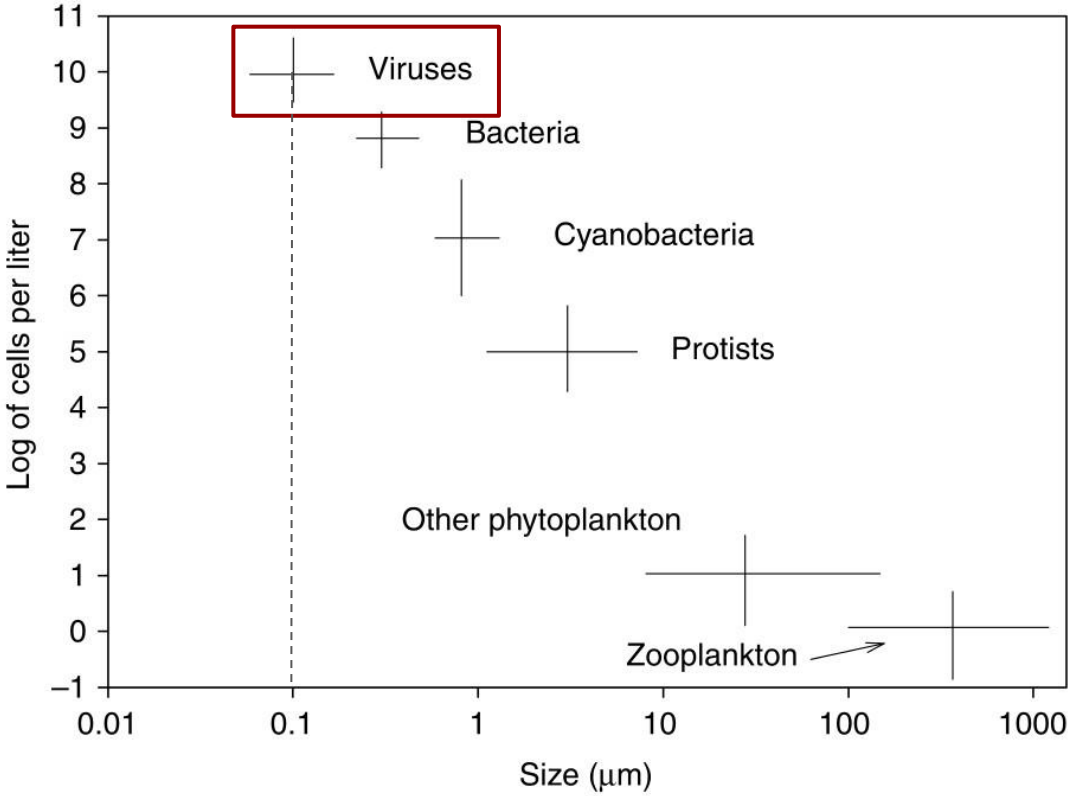
Huge potential to **influence diversity, evolution and biogeochemistry**

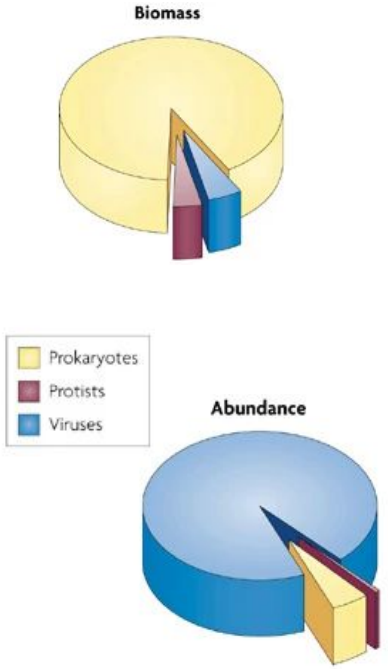
Virus family	Morphology of virion	Size of virion (nm)[a]	Marine host(s)
Double-stranded RNA viruses			
Birnaviridae	Round, icosahedral	60	Molluscs, fish
Cystoviridae	Icosahedral with lipid coat	60–75	Bacteria
Reoviridae	Icosahedral, some with spikes	50–80	Crustaceans, molluscs, fish, protists
Totiviridae	Round, icosahedral	30–45	Protists
Single-stranded RNA viruses (positive sense)			
Caliciviridae	Round, icosahedral	35–40	Fish, mammals
Coronaviridae	Rod-shaped with projections	200 × 42	Crustaceans, fish, seabirds
Dicistroviridae	Round, icosahedral	30	Crustaceans
Leviviridae	Round, icosahedral	26	Bacteria
Marnaviridae	Round, icosahedral	25	Algae
Nodaviridae	Round, icosahedral	30	Fish
Picornaviridae	Round, icosahedral	27–30	Algae, crustaceans, thraustochytrids, protists (?), mammals
Togaviridae	Round, with outer fringe	66	Fish
Single-stranded RNA viruses (negative sense)			
Bunyaviridae	Round, enveloped	80–120	Crustaceans
Orthomyxoviridae	Round, with spikes	80–120	Fish, mammals, seabirds
Paramyxoviridae	Various, mainly enveloped, filamentous	60–300 × 1000	Mammals
Rhabdoviridae	Bullet-shaped with projections	45–100 × 100–430	Fish

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Viral Abundance VS Size

There are more than 10^{30} viruses in the oceans



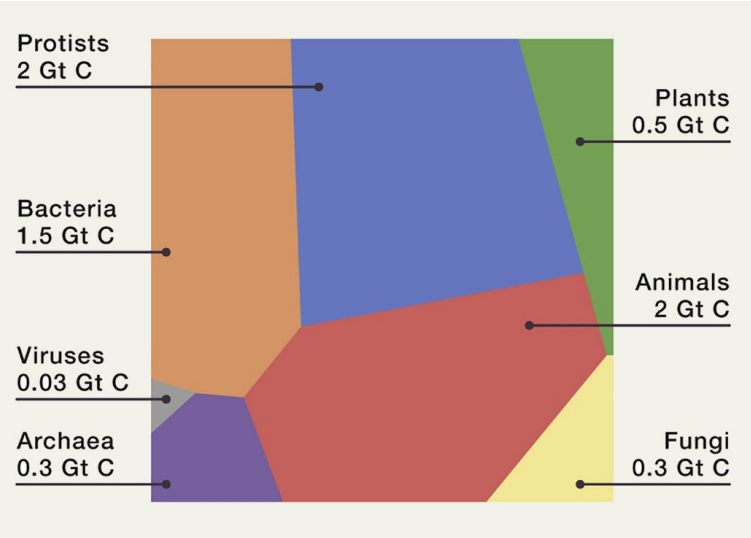


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10.1038/nrmicro1750

The Biomass Composition Of The Oceans

In contrast to their domination on land, plants (green algae and seagrasses) account for less than 10% of the total biomass in the ocean (~6 GtC), whereas animals correspond to ~33%.
Viruses dominate the ocean in terms of number, with a virus-to-microbial cell ratio of 10:1, but constitute only ~0.5% of the total biomass.
Marine microbes, including bacteria, archaea, protists, fungi and viruses, represent ~60% of total marine biomass, compared with ~18% on land.
These reported estimates for global marine biomass have been filtered to include only marine organisms. Therefore, all the microbial biomass, bacteria and archaea, living in the deep subsurface environment (deep aquifers and the ocean's crust) has not been even represented.



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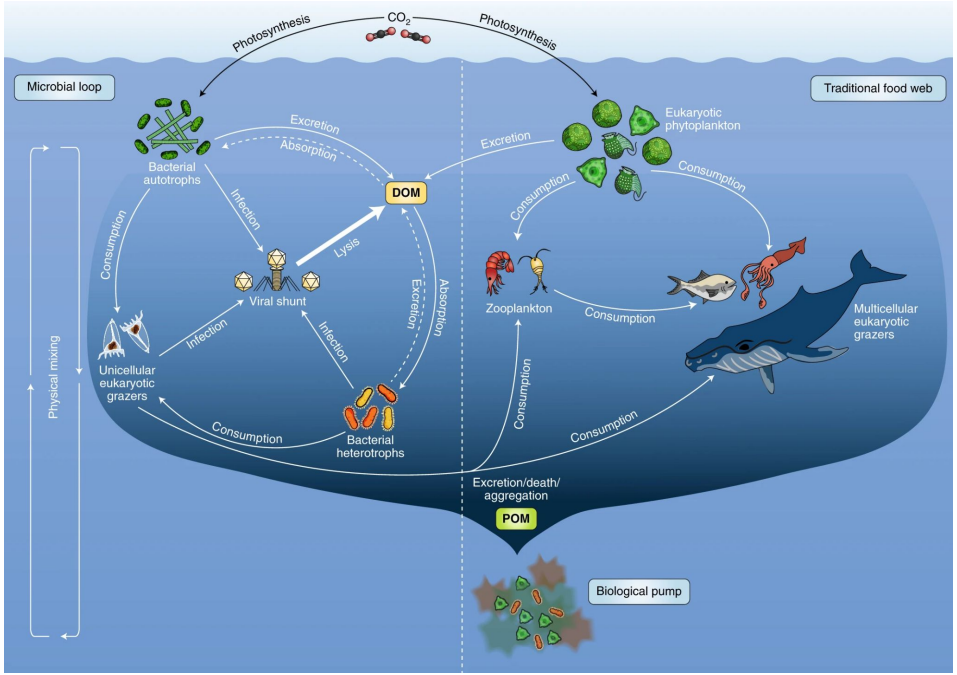
10.1016/j.cell.2019.11.018

Virus-Host Dynamics

Between 20-30% of marine bacterial cells are thought to be infected by phages at any given time, with the viral shunt releasing up to 3 GtC into the oceans per year. An estimated 10^{28} viral infections occur per day in the oceans. To put this into perspective, in every blink of an eye (0.3 seconds), the number of viral infections in the ocean (3×10^{22}) equals the number of stars in the universe.

In the time it took you to read that sentence, ~10^{24} viral infections occurred in the oceans.

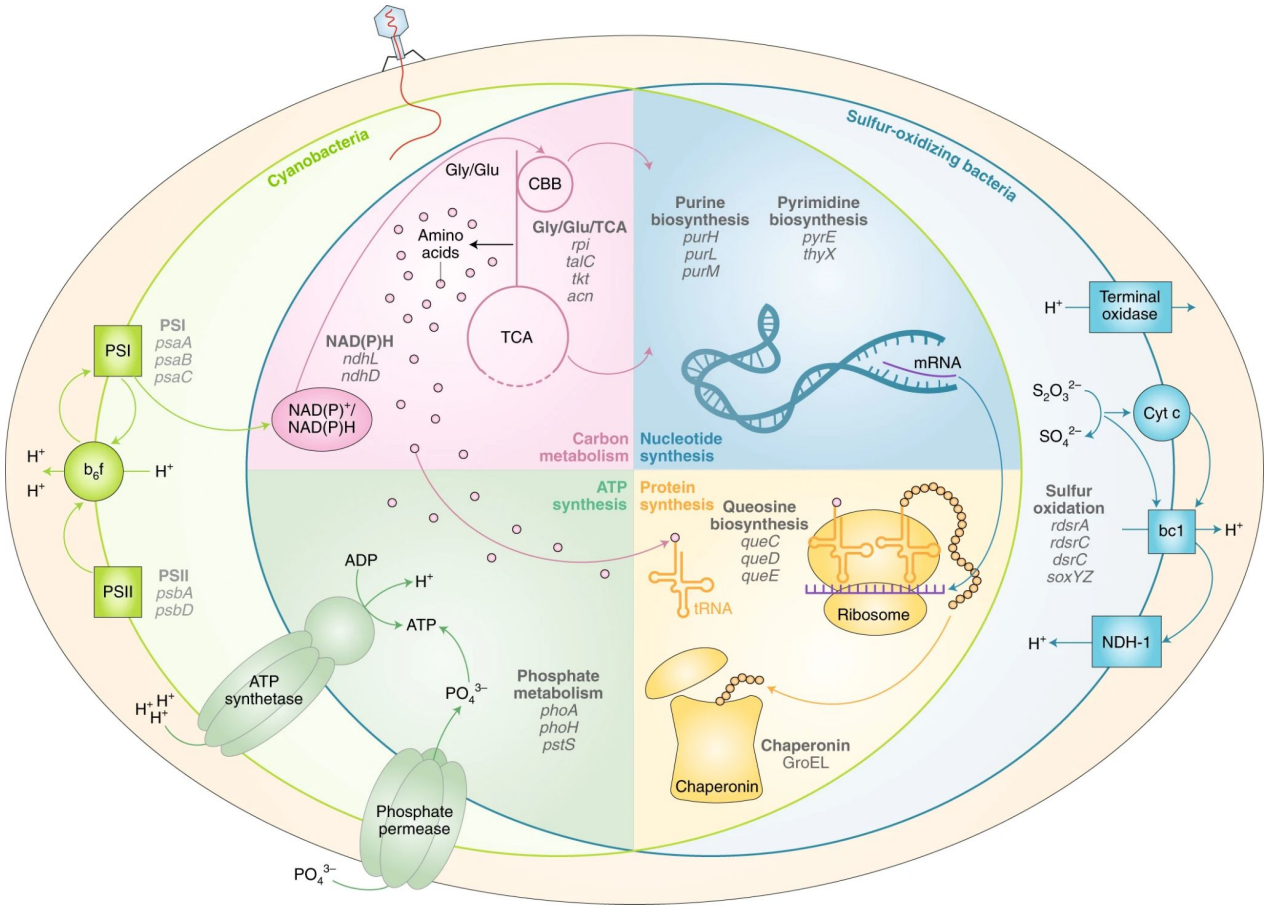
These numbers emphasize the importance of phage lysis as a major force in the rapid recycling of carbon and nutrients within the oceans.



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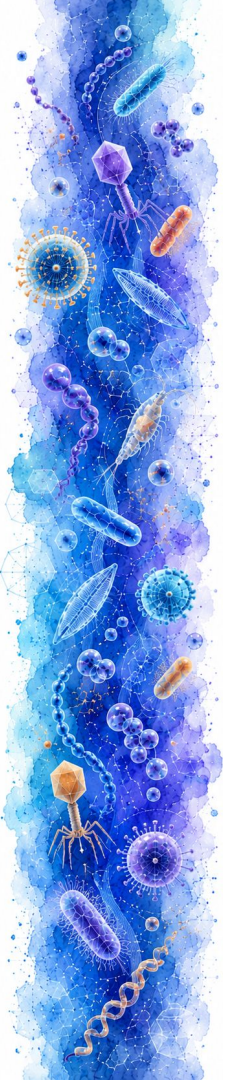
https://doi.org/10.1038/s41564-018-0166-y

Virus-Host Dynamics



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https://doi.org/10.1038/s41564-018-0166-y



This Week Read

Caron DA, Countway PD, Jones AC, Kim DY, Schnetzer A. Marine protistan diversity. Ann Rev Mar Sci. 2012;4:467-493. doi:10.1146/annurev-marine-120709-142802

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